XML Interface
Ticket System – Handshake.Logic

Confidential technical Information to Handshake Partners

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SKIDATA KUDELSKI GROUP 4.0
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Version History

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31.10.2001	1.00		THMI	Final extensions, released to TS partners
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1. Glossary

Term	Description
Blacklist	List of blocked tickets
Coding	Coding technology and data carriers
ET	E-Ticket, also used as abbreviation for the SKIDATA e-ticketing service
FDE	First Day Entry
FE	First Entry
FFU	For future use
GDPR	General Data Protection Regulation
HSH	Handshake.Logic
HSHID	Unique Handshake System Id
MultiDC	Multiple Data Carrier (multiple data carriers assigned to 1 permission)
POS	Point of Sale
RF	Radio Frequency Technology
TEG	Ticketing-System – E-Ticket – Gateway
TS	Ticketing System
TSID	Unique Ticketing System Id
TSProperty	Type of Information sent by Ticketing System (EVENT, AREA, …)
UPID	Unique Permission Id
UTID	Unique Ticket ID provided by the Ticketing System. The UTID describes the electronically coded information of the issued ticket, which guarantees uniqueness of the ticket within HSH. Can be the whole electronically coded information or a part of it.
Whitelist	List of allowed tickets
XML	Extended Mark-up Language

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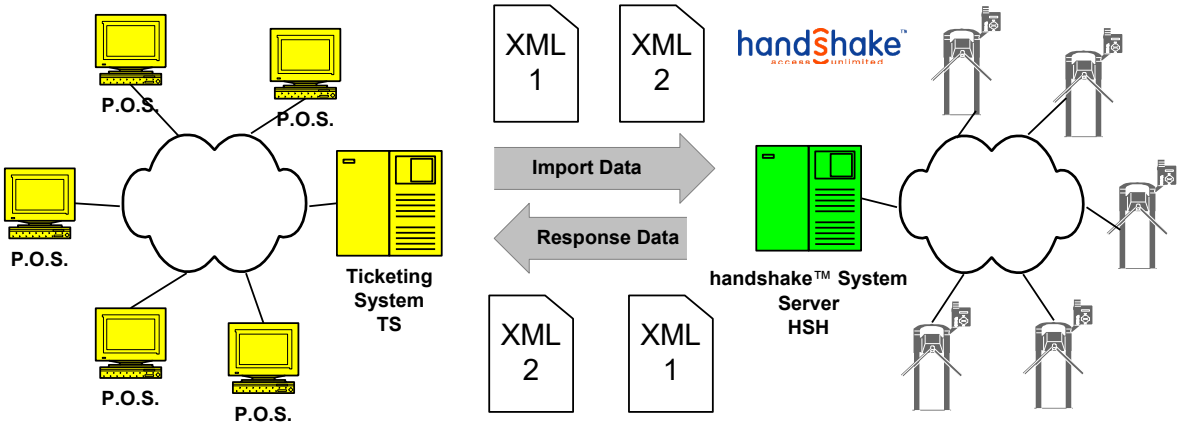
2. General

This document describes the possibilities using the XML interface connecting Handshake.Logic (HSH) and a Ticketing System (TS). Handshake.Logic is an access system designed to interface seamlessly with existing ticket systems.

2.1 Interface between TS and HSH

The following picture shows the standard configuration using HSH linked to TS. The TS may send the information via various connections (LAN, WAN, etc.) depending on the capabilities of the TS.

In most cases the TS Server communicates with the HSH Server, but it is also possible that the POS devices communicate directly with the HSH Server.



The interface between Handshake.Logic (the access system) and the ticketing system (the ticket information provider) is based on two possible channels:

2.1.1 Receiving data (from ticketing system)

- **Ticketing system properties** (TSProperties) – configuration data describing the event, sectors (areas), ticket types etc. are used for the access configuration in the Handshake.Logic.
- **Whitelist information** – contains information about valid tickets and assigned properties.
- **Blacklist information** – contains information about invalid (blocked) tickets
- **Inquiry** – request of statistical data (transactions, counters, …), permission details, import details.

2.1.2 Sending transaction data (to ticketing system)

HSH replies on requests received from ticket system.
The ticketing system can

- Inquire the usage information of tickets
- Inquire the entire usage statistic (transactions)
- Inquire ticket details (permissions, points, days, time …)
- Inquire counters (current number of visitors)
- Inquire imports

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2.2 Communication

When the physical link is established between TS and HSH the data exchange is started. Each telegram containing a TSData element from the TS gets an answer from HSH with an HSHData element. If the XML-telegram was sent without errors the HSHData element simply contains the Header element. In case of error an Error element describes the problem at the HSH side.

For the file interface the behavior of HSH responses can be configured within the HSH Explorer. It's possible to configure HSH that no answer is sent (HSHData element), an answer is sent for every import or the answer is sent only in case of errors. This configuration at the HSH side also depends on the capabilities of the TS.
The socket and the web service interface always send a response.

2.2.1 Example standard communication without error

Data from TS

```
<TSData>
     <Header>
        <Version>HSHIF25</Version>
        <Issuer>1</Issuer>
        <Receiver>1</Receiver>
        <ID>1</ID>
     </Header>
     <TSProperty Type="EVENT">
        <Action>I</Action>
        <ID>3003</ID>
        <Name>Elton John Concert</Name>
        <Arguments>
           <Place>Vienna concert hall</Place>
           <FromDate>2018-10-26</FromDate>
           <FromTime>20:00</FromTime>
           <ToDate>2018-10-26</ToDate>
           <ToTime>23:30</ToTime>
        </Arguments>
     </TSProperty>
</TSData>
```

Answer from HSH

```
<HSHData>
     <Header>
        <Version>HSHIF25</Version>
        <Issuer>1</Issuer>
        <Receiver>1</Receiver>
        <ID>1</ID>
     </Header>
</HSHData>
```


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2.2.2 Example standard communication with error

Data from TS

```
<TSData>
     <Header>
        <Version>HSHIF25</Version>
        <Issuer>1</Issuer>
        <Receiver>1</Receiver>
        <ID>1</ID>
     </Header>
     <TSProperty Type="EVENT">
        <Action>X</Action>
        <ID>3003</ID>
        <Name>Elton John Concert</Name>
        <Arguments>
           <Place>Vienna concert hall</Place>
           <FromDate>2018-10-26</FromDate>
           <FromTime>20:00</FromTime>
           <ToDate>2018-10-26</ToDate>
           <ToTime>23:30</ToTime>
        </Arguments>
     </TSProperty>
</TSData>
```

Answer from HSH

```
<HSHData>
     <Header>
        <Version>HSHIF25</Version>
        <Issuer>1</Issuer>
        <Receiver>1</Receiver>
        <ID>1</ID>
     </Header>
     <Error>
        <Type>E</Type >
        <Number>20000</Number>
        <Description>Invalid Action Type</Description>
        <TSProperty Type="EVENT">
           <Action>X</Action>
           <ID>3003</ID>
           <Name>Elton John Concert</Name>
           <Arguments>
                <Place>Vienna concert hall</Place>
                <FromDate>2018-10-26</FromDate>
                <FromTime>20:00</FromTime>
                <ToDate>2018-10-26</ToDate>
                <ToTime>23:30</ToTime>
           </Arguments>
        </TSProperty>
     </Error>
</HSHData>
```


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2.3 Process overview

2.3.1 System setup

During the system setup all configuration data (TSProperties) must be sent to HSH. Configuration data are e.g. AREA, PERSONCATEGORY or TICKETTYPE. If the configuration is changed the configuration data must be transferred to HSH again.

2.3.2 Event setup

If a system is based on events, every time a new event is configured, the event details must be sent to HSH. This must be done right in time. The event information is used by HSH to setup the event details.

If the installed system uses season passes, normally with the event information the WhitelistRecords for the season passes are also provided to HSH.

2.3.3 Sales process

When a ticket is produced a WhitelistRecord is generated and transferred to HSH. The WhitelistRecord should be generated only if the production of the ticket was successful.

2.3.4 Cancelation process

If a ticket is canceled, two possible situations must be considered:

- The ticket is canceled after the production and the electronically coded information on the ticket is invalid. In this case it is enough to delete the previously sent WhitelistRecord.
- The ticket is canceled after the production, the electronically coded information on the ticket is still valid. A BlacklistRecord must be created to deny the usage of the ticket. To be able to display proper information at the checkpoint for the canceled ticket, it is recommended to send a BlacklistRecord additionally to the WhitelistRecord.

2.3.5 Inquiry process

Depending on the capabilities of the TS, the POS should be able to get information about the usage of a ticket, before it is canceled. The Inquiry enables the TS to request ticket usage information from HSH.

3. Data definition

All possible attributes and elements are listed in this document. If an XML attribute or element is defined as mandatory, this attribute or element must be part of the XML telegram. Each attribute has a maximum length definition. The length shows the maximum value, the value though may be shorter than the specified maximum length.

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3.1 Used XML data types

Hint: HSH supports the standard XML data types.

XML Data Types	Description
boolean	Specifies a Boolean value of either True or False. E.g.: 0, 1 (1=true)
dateTime	Specifies a date in a subset of the ISO 8601 format, with optional time and optional time zone information. Fractional seconds can be as precise as nanoseconds.
date	Specifies a date in a subset of the ISO 8601 format.
time	Specifies a time in a subset of the ISO 8601 format.
i4	Specifies a four-byte integer (Dword) with optional sign (-2147483648 to 2147483647).
i8	Specifies an eight-byte integer (Qword) with optional sign (-9223372036854775808 to 9223372036854775807).
string	Specifies a string. E.g.: This is a string

The data sent in the XML telegram must contain the same values as coded on the ticket (e.g. „serial number" …)!

Note**:** Leading zeros or leading blanks coded on the ticket must be also sent within the XML telegram.

This is necessary because the data stored on the ticket is compared to the data sent in the XML telegram. The information on the ticket must match the Whitelist information for a successful ticket check.

3.1.1 Date-Time zone information

Any date-time object transferred to and from Handshake server shall contain unambiguous date-time information.
From Handshake version 24.0 it is also possible to send a date-time information either in UTC or it contains time-zone information.
If time zone is missing, the local time of the Handshake server is assumed.

The interface configurations that are persisted in the database are containing an XML attribute **UseTimeZone**. Its value is 0 by default:

```
<Configuration>
```


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```
  <Transport Type="WebService"
Class="SkiData.Handshake.Import.Transports.WebServiceTransport"
Assembly="SDImportHandlers.dll">
    <Parameters Port="5007" UseAuthentication="0" UseHttps="0"
ArchiveDirectory="C:\Importer\Photo\%Y%M%D" WriteArchive="1" Timeout="300"
/>
  </Transport>
  <Converter Type="XML"
Class="SkiData.Handshake.Import.Converters.XmlConverter"
Assembly="SDImportHandlers.dll">
    <Parameters Version="HSHIF25" MapAddRel="0" MonitoringInImporter="1"
UseTimeZone="0"/>
  </Converter>
</Configuration>
```

If date-time objects transferred from Handshake server back to the TS should include date time information as well, **UseTimeZone** must be set to 1. Currently it cannot be configured via graphical user interface, the value must be updated manually in the database for the concerning interface.

3.2 XML data structure

The following table shows the structure of the used XML-Data, for details please see chapter 4 Overview of XML elements.

3.2.1 Data from TS to HSH

HSH can process different data elements sent by the TS. These data elements are for TSProperties, Whitelist, Blacklist and Inquiry.

Data elements contain action information how to process the sent data.

- **Insert** – Inserts the information into HSH
- **Delete** – Deletes the information from HSH
- **Update** – HSH updates existing information. Used to change incorrect information and must not include key values. If the key element must be updated, the existing record must be deleted and inserted again, key elements are defined for each property.
- **Anonymize** – Anonymizes personal information (GDRP)

XML-Data from TS	Description
TSData	See TSData element
Header	See Header element
Version	
Issuer	
Receiver	
ID	
TSProperty	See TSProperty element
Type	

Internal

XML-Data from TS	Description
Expire	
From	
To	
Action	
ID	
Name	Note: if type is EVENT the name will be cut off at 30 characters.
Arguments	See Arguments element
Place	
FromDate	
FromTime	
ToDate	
ToTime	
TSProperty	See TSProperty element
WhitelistRecord	See WhitelistRecord element
Expire	
Action	
UTID	
Coding	
Permission	See Permission element
UPID	
TSProperty	See TSProperty element
BlacklistRecord	See BlacklistRecord element
Expire	
Action	
WhitelistRecord	See WhitelistRecord element
Permission	See Permission element
UPID	
TSProperty	See TSProperty element
BlockingType	
BlockingReason	

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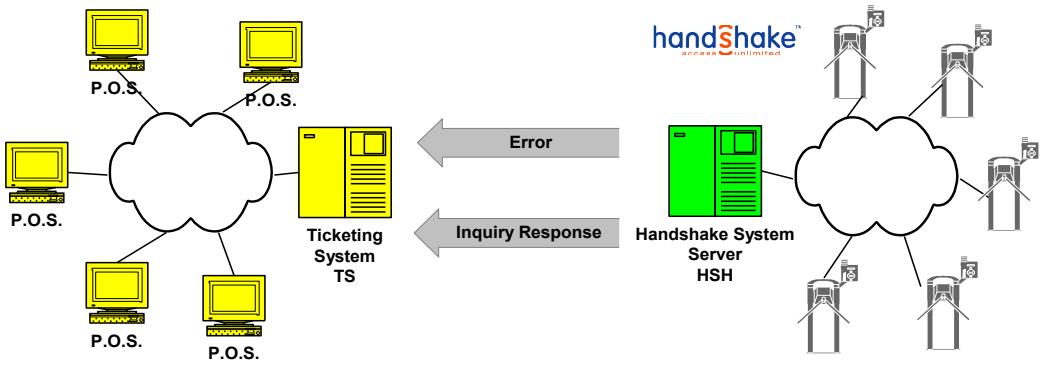
XML-Data from TS	Description
From	
To	
DisplayMessage	
Comment	
Inquiry	See Inquiry element
Type	
Option	
Status	All … All permissions Valid … Valid permissions Blocked … Blocked permissions
From	
To	
LastTransactionId	
UTID	
Coding	
Permission	See Permission element
UPID	
TSProperty	See TSProperty element

Internal

3.2.2 Data from HSH to TS

After processing the XML telegram HSH sends import information. The import process generates a response telegram, with information about the import result. In case of a successful import an OK telegram is sent and in case of an error telegram is sent.
The Inquiry is a special data type, used by the TS to inquire data from HSH. As response HSH answers with an Inquiry element. In case the requested data is unknown in HSH (e.g. missing Whitelist information) an InquiryError element is generated.



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XML-Data from HSH	Description
HSHData	See HSHData element
Header	See Header element
Version	
Issuer	
Receiver	
ID	
Inquiry	See Inquiry element
Type	
From	
To	
Timestamp	
Option	
LastTransactionId	
NotComplete	
Ticket	See Ticket element
UTID	
Coding	
RawData	See RawData element
Permission	See Permission element
UPID	
Blocked	No … Valid Permission Yes … Blocked Permission
TSProperty	See TSProperty element
TicketUsed	
LastDate	
Transaction	See Transaction element
TransactionId	
Date	
BelongsTo	
Status	
Validated	

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XML-Data from HSH	Description
Releases	
PermissionType	
CurrentValue	
UsedValue	
Venue	
Area	
Gate	
Checkpoint	
ExtendedCheck	See ExtendedCheck element
Time	See Time element
InquiryError	See InquiryError element
NumberOfUPID	
Counter	See Counter element
Id	
Name	
Type	
Location	See Location element
Id	
Name	
Type	
Entries	
Exits	
UpperLimit	
LowerLimit	
Locked	
Totals	See Totals element
WhitelistRecord	
BlacklistRecord	
Error	See Error element
Type	
Number	

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XML-Data from HSH	Description
Description	
TSProperty	See TSProperty element
WhitelistRecord	See WhitelistRecord element
BlacklistRecord	See BlacklistRecord element
InquiryError	See InquiryError element
No	
Text	

Internal

4. Overview of XML elements

4.1 TSData element

TSData is the root element for all telegrams sent from the TS.

Name	Xml Attribute/Element	Mandatory	Data Type	Remark
Header	Element	Yes	Elem.	See Header element
TSProperty	Element		Elem.	See TSProperty element
WhitelistRecord	Element		Elem.	See WhitelistRecord element
BlacklistRecord	Element		Elem.	See BlacklistRecord element
Inquiry	Element		Elem.	See Inquiry element

4.2 HSHData element

HSHData is the root element for all response telegrams sent by HSH. In case of an error HSHData contains an Error element with error Information.

Name	Xml Attribute/Element	Mandatory	Data Type	Remark
Header	Element	Yes	Elem.	See Header element
Error	Element		Elem.	See Error element
Inquiry	Element		Elem.	See Inquiry element

4.3 Header element

The Header element contains the partner information (sender and receiver), the HSH interface version and the ID number.

Note: The ID must be unique for a time period of at least 24 hours. We recommend that the ID is increased by 1. This will help to find missing data telegrams in case of error.

The XML telegram must contain exactly **one** Header element!

The values for Issuer and Receiver are defined by SKIDATA, the values must be configurable.

Note: The Issuer in the XML-Header and the Issuer in the barcode have different meaning and must be distinguished.

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Name	Attribute/Element	Mand.	Data Type	Max. Length	Remark	Example
Version	Element	Yes	string	100	Interface Version Fix keyword „HSHIF25" defined by HSH	HSHIF25
Issuer	Element	Yes	i4	11	Sender of a telegram the Issuer is set in the HSH configuration	2
Receiver	Element		i4	11	Receiver of a telegram This keyword is Important if HSH is used by more than 1 client, the Receiver defines for which client the telegram is sent. Receiver is set in the HSH configuration	1 … Club 1 2 … Club 2
ID	Element		i4	11	Unique ID for XML-Data generated by TS, useful in case of error (e.g.: to find missing data)	1

4.3.1 Header sample

```
<Header>
    <Version>HSHIF25</Version>
    <Issuer>2</Issuer>
    <Receiver>1</Receiver>
    <ID>1</ID>
</Header>
```

4.4 Error element

The Error element contains information about incorrect and irregular situations during the import process. It may be necessary to resend the telegram or send additional information. Most of the time, it will be necessary to correct the situation before continuing sending data.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Type	Attribute		Yes	string	1	S … critical Error (stop) import is aborted E … error W … warning	W
Number	Element	Yes	Yes	i4	11	Error number	30001
Description	Element			string	100	Error description	Already Existing Property

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TSProperty	Element					See TSProperty element	
WhitelistRecord	Element					See WhitelistRecord element	
BlacklistRecord	Element					See BlacklistRecord element	

4.4.1 Error sample

```
<Error Type="E">
     <Number>20000</Number>
     <Description>Invalid Action Type</Description>
     <TSProperty Type="PERSONCATEGORY">
        <Action>X</Action>
        <ID>20</ID>
        <Name>Adult</Name>
     </TSProperty>
</Error>
```

4.4.2 TEG web link result warning

It the TEG configures e-ticket creation with delivery mode ‘Return web link’ then the link to the e-ticket is reported back to the TS by adding a warning (error number 32001).
The link to the e-ticket is transferred in the <Description> element.

```
<Error Type="W">
     <Number>32001</Number>
     <Description>https://...</Description>
     <WhitelistRecord Expire="2018-12-21T01:02:03">
        <Action>I</Action>
        <UTID>29649183693</UTID>
        <Coding>262</Coding>
        <Permission MultiDC="false">
           <UPID>1</UPID>
        </Permission>
     </WhitelistRecord>
</Error>
```

4.4.3 List of error numbers

The following tables show all available error messages and warnings.

4.4.3.1 Severe errors

Error Number	Error Type	Error Description	Remark
10000	S	Unknown Record Type	XML Data does not match HSH interface.
10001	S	Unknown Interface Version	Header element field version contains unknown version information.
10002	S	Unknown Ticketing Server	Must be configured in the HSH system before starting import.
10003	S	Duplicate ID	Header element field ID must be unique at least for 24 hours (happens when record is sent twice).

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Error Number	Error Type	Error Description	Remark
10004	S	Invalid XML-Record	Error in XML-Record.
11000	S	HSH internal error	General error occurred within the HSH interface process.
11001	S	HSH database error	Error concerning HSH Database.
12000(1)	S	Test mode active	TEG is running in test mode (i.e. not executing HSH and ET operations)
12001(1)	S	HSH not reachable	TEG can't connect to HSH
12002(1)	S	Empty HSH response	HSH responded to a request with an empty response.

(1)…Only occur if the TEG is active.

4.4.3.2 Errors

Error Number	Error Type	Error Description	Remark
20000	E	Invalid Action Type	Supported types are I (Insert), D (Delete), U (Update) and A (Anonymize).
20001	E	Unknown XML-element	Unknown XML element was sent, HSH don't know how to handle.
20003	E	Invalid Data Type	Value is invalid for used data type; invalid format e.g. for date values.
20004	E	Missing Mandatory Field	Mandatory information is missing.
20005	E	Missing Key Column	When using action "U" for update all key columns must be provided.
20006	E	Unknown Event ID	Event ID used before sending area specific event property.
20007	E	Unknown Area ID	Area ID used before sending area specific area property.
20008	E	Unknown Person ID	Person ID used before sending specific person category.
20009	E	Invalid Coding	Invalid value not listed in „Coding definition Table" was used.
20010	E	Invalid Blocking Type	Unspecified value was used for „Blocking Type".
20011	E	Invalid Blocking Reason	Unspecified value was used for „Blocking Reason".

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20012	E	TSProperty already in Use	TSProperty cannot be deleted because it is already used.
20013	E	Whitelist Update not Possible	Whitelist Update is not possible because used TSProperty can be assigned to the Whitelist more than once.
20014	E	Non Existing Permission	The specified permission (incl. its attributes, i.e. multi-data carrier) in the specified context (indicated by UTID and Coding) could not be found.
20015	E	Non Existing PHOTO ID	Reference photo specified by PHOTO ID doesn't exist.
20016	E	Invalid attribute value	The specified inquiry type is unknown.
20017	E	Unknown ticket	Ticket could not be found.
20018	E	Non existing RequestId	If the value of Action element is A, a RequestId is mandatory.
20019	E	Missing Mandatory Field To	Mandatory information for field "To" is missing.
20020	E	Missing Mandatory Field BlockingReason	Missing value for element BlockingReason.
20021	E	Missing Mandatory Field BlockingType	Missing value for element BlockingType.
20022	E	Coding Element too short	Value for element coding is too short - min lenght is 1
20023	E	UTId element too short	Value for element UTId is too short - min lenght is 1
20024	E	Data anonymization not enabled	Data anonymization not enabled
20025	E	Invalid combination of attributes	MultiDC and HSHUPId attributes must not be used at the same time.
20026	E	Unknown permission	Permission could not be found.
22000(1)	E	Indeterminate ticket definition	The e-ticket definition can't be determined.
22001(1)	E	Indeterminate delivery method	The e-ticket delivery method can't be determined.
22002(1)	E	Missing email address	The e-ticket recipient (customer e-mail address) is missing.
22003(1)	E	Indeterminate service partner	The e-ticket service partner can't be determined.

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22004[1]	E	Indeterminate identification type	The e-ticket identification type can't be determined.
22005[1]	E	Missing encryption key	The (NFC) e-ticket encryption key can't be determined.
22006[1]	E	Indeterminate data carrier content	The e-ticket data carrier content can't be determined.
22007[1]	E	Missing ticket definition	The e-ticket definition (determined from TS data) doesn't exist in the ET tenant.
22008[1]	E	No value for mandatory property	The value of a mandatory e-ticket property is missing in the TS data.
22009[1]	E	E-Ticket operation no results	An ET operation (e-ticket create, update, ticket-info) returned an empty result.
22010[1]	E	E-Ticket operation server error	An ET operation (e-ticket create, update, ticket-info) failed with a server error.
22011[1]	E	E-Ticket operation exception.	An ET operation (e-ticket create, update, ticket-info) failed with an exception.
22012[1]	E	E-Ticket not found	The ET operation ticket-info didn't find the existing ticket.
22013[1]	E	Unsupported coding	The TS ticket coding is not supported / can't be used to create an e-ticket.
22014[1]	E	No TS properties	The TS data (permission) has no TS properties

[1]…Only occur if the TEG is active.

4.4.3.3 Warnings

Error Number	Error Type	Error Description	Remark
30000	W	Non-Existing TSProperty	Delete or update action on non-existing TSProperty.
30001	W	Already Existing TSProperty	TSProperty already defined within HSH, Insert produces duplicate key (happens when property is sent twice)
30002	W	Update instead of Insert, because TSPropertyType already exists	TSPropertyType already exists for WhitelistRecord, Insert becomes Update because this TSPropertyType can only be assigned once.
30003	W	Delete, record does not exist	The record that should be deleted does not exist.
32000[1]	W	Skipped by configuration	The e-ticket isn't created because the e-ticket definition is indetermined. The TEG configuration parameter

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			TicketDefinitionProperty.SkipEtIfValueNotMapped specifies to raise a warning instead of the 22000 error.
32001[(1)]	W	Web link result	The e-ticket has been created with delivery method ReturnWebLink. The web link to the e-ticket is available in the Error.Description.
32002[(1)]	W	Update – not create	The e-ticket wasn't created (it already existed) – so it was updated.

[(1)]…Only occur if the TEG is active.

4.4.4 TEG error elements

If activated, the TEG adds Error elements to the TSData response, containing information about incorrect and irregular situations during the e-ticket operation(s).

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4.5 TSProperty element

The TSProperties are imported and mapped to HSProperties within the HSH configuration. HSProperties are used to set up the access configuration (permit or deny access at the checkpoints) in HSH.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Type	Attribute	Yes	Yes	string	30	e.g. EVENT AREA PERSONCATEGORY VALIDITY	EVENT
From	Attribute			dateTime	19	Begin date (Optionally includes time zone information)	2018-12-31T01:33:00 (2018-12-31T01:33:00+04:00)
To	Attribute			dateTime	19	End date (Optionally includes time zone information)	2018-12-31T02:33:00 (2018-12-31T02:33:00+04:00)
Expire	Attribute			dateTime	19	Date when property can be archived (Optionally includes time zone information)	2018-12-31T00:00:00 (2018-12-31T00:00:00+04:00)
Action	Element		Yes	string	1	I… insert D … delete U …update A … anonymize	D
ID	Element	Yes	Yes	string	100	Unique number for TSProperty generated by the TS (unique by Type)	123
Name	Element		Yes	string	100	Name of TSProperty	Elton John concert
Arguments	Element			Elem.		E.g.: Used with EVENT Property	

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						See Arguments element	
TSProperty	Element			Elem.		If a TSProperty depends on another e.g. PersonCategory „Fan” only valid for Event „Elton John Concert”	

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4.5.1 TSProperty sample

```
<TSProperty Type="EVENT">
     <Action>I</Action>
     <ID>3003</ID>
     <Name>Elton John Concert</Name>
     <Arguments>
        <Place>Vienna concert hall</Place>
        <FromDate>2018-10-26</FromDate>
        <FromTime>20:00</FromTime>
        <ToDate>2018-10-26</ToDate>
        <ToTime>23:30</ToTime>
     </Arguments>
</TSProperty>
```

4.6 Arguments element

Arguments can be used to provide additional information for events. The arguments element contains place, date and time information for an event.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Place	Element			string	100	Place of event	Vienna concert hall
FromDate	Element			date	10	Begin date of event	2018-10-26
FromTime	Element			time	5	Begin time of event	20:00
ToDate	Element			date	10	End date of event	2018-10-26
ToTime	Element			time	5	End time of event	23:30
FromDateTime	Element			datetime in ISO format with time zone	25	End datetime of an event	2018-10-26T20:00:00+04:00 or 2018-10-26T20:00:00Z
ToDateTime	Element			datetime in ISO format with time zone	25	End datetime of an event	2018-10-26T23:30:00+04:00 or 2018-10-26T23:30:00Z

4.6.1 Arguments sample

```
<Arguments>
     <Place>Globen</Place>
```


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```
    <FromDate>2018-10-26</FromDate>
    <FromTime>20:00</FromTime>
    <ToDate>2018-10-26</ToDate>
    <ToTime>23:30</ToTime>
    <FromDateTime> 2018-10-26T20:00:00+04:00</FromDateTime> (optional)
    <ToDateTime>2018-10-26T23:30+04:00</ToDateTime> (optional)

</Arguments>
```

4.6.2 Date and time format for event setup

If the elements FromDate, FromTime, ToDate and ToTime and the optional elements FromDateTime and ToDatetime should be used to setup the event, date and time and optional datetime must be provided in one of the following formats.

Date:
yyyy-mm-dd
yyyymmdd

Time:
hh:mm
hhmm

DateTime:
yyyy-mm-ddThh:mm:ssK
yyyy-mm-ddThh:mm:ssK

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4.7 Standard TSProperties

The following standard TSProperties that exist in HSH can be extended dynamically as required. To be able to send a new TSProperty type to HSH, the new property type must be configured within HSH.

4.8 TSProperty EVENT

The EVENT property shows information about events (e.g. matches). HSH does not make a difference between event types e.g. performances, games, matches, seasons etc.

Note: If type is EVENT the name will be cut off at 30 characters.

4.9 TSProperty AREA

To make a difference for tickets to be allowed to enter one or more areas, the AREA property is used. This property is used to show the names used for areas in TS within HSH and helps to make the Whitelist human readable.

4.10 TSProperty PERSONCATEGORY

The PERSONCATEGORY is used to show the names used for person categories in TS within HSH (e.g. Adult ticket, Child ticket) and to be able to configure access permissions and restrictions for different person categories.

4.11 TSProperty TICKETTYPE

Property TICKETTYPE is used to distinct day tickets, season passes, point tickets, group tickets or time tickets.

4.12 TSProperty ADDRELEASES

A standard passage is one release of the barrier. The Property ADDRELEASES informs HSH that the turnstile must be opened not only for the owner of the ticket but additionally as many times as defined with the Element „ID”. If the „ID” has the value 1 this means two entries are allowed with the specified ticket. The TSProperty ADDRELEASES is designed to be used e.g. for escort people of a handicapped person.

4.13 TSProperty VALIDITY

The VALIDITY property consists of the two attributes **From** and **To**. The attribute **From** specifies the starting date and the attribute **To** the end date of a VALIDITY. At least one of the two attributes must be provided. If both attributes are used, the VALIDITY starts at the **From** date and ends at the **To** date. When only **From** is used, the VALIDITY begins with the From date with open end. Using only the **To** date, the VALIDITY starts at the current date and ends at the To date.

It’s important that the VALIDITY attribute contains proper time information. It is not enough to send only the date. E.g.: From="2018-01-01T00:00:00" To="2018-01-01T00:00:00" as the start and the end is at the same time it is never valid. The VALIDITY must be sent with the WhitelistRecord (see WhitelistRecord samples).

Internal

4.13.1 VALIDITY sample

```
<TSProperty Type="VALIDITY" From="2018-01-01T00:00:00" To="2018-12-31T23:59:00" />
```

With time zone information:

```
<TSProperty Type="VALIDITY" From="2018-01-01T00:00:00+04:00" To="2018-12-31T23:59:00+04:00" />
```

4.14 TSProperty DAYS

This TSProperty assigns the defined number of days to a day ticket (e.g.: 1 day, 5 days, etc.). With the TSProperty DAYS the ticketing system defines how many days a ticket is valid. The unit of the TSProperty is days. The TSProperty DAYS can only be used as part of the Whitelist.
If additional days must be added to the existing day value, the property DAYS is used with a + sign in front of the ID.
If the number of days needs to be reduced, the property DAYS is used with a - sign in front of the ID.

4.14.1 DAYS sample

```
<TSProperty Type="DAYS">
     <ID>1</ID>
</TSProperty>
```

4.14.2 Add additional DAYS sample

```
<TSProperty Type="DAYS">
     <ID>+2</ID>
</TSProperty>
```

4.14.3 Reduce number of DAYS sample

```
<TSProperty Type="DAYS">
     <ID>-2</ID>
</TSProperty>
```

4.14.4 Extension maximum number of days

It is also possible to define beside the number of valid days also maximum permissible number of days (max days). The TSProperty DAYS can be used to deliver both values separated by slash: “x/y”. In that case the first value are x days and the second are y maximum days.
If the period of days (max days) since first use of ticket is expired, all remaining valid days of the day ticket which have not been consumed will expire automatically and therefore the permission is not valid anymore.

4.14.4.1 DAYS with extension maximum number of days sample

```
<TSProperty Type="DAYS">
     <ID>2/7</ID>
</TSProperty>
```

This means that the permission is valid for 2 days and it must be used within a period of 7 days (max days) from the first use.

4.15 TSProperty POINTS

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This TSProperty is used for giving point tickets a value (e.g.: 10 points, 50 points, etc.). With the TSProperty POINTS the Ticketing System defines how many points are on the specified ticket. The TSProperty POINTS can only be used as part of the Whitelist.

If additional points need to be added to the existing point value, the property POINTS is used with a + sign in front of the ID.

If the number of points needs to be reduced, the property POINTS is used with a - sign in front of the ID.

4.15.1 POINTS sample

```
<TSProperty Type="POINTS">
     <ID>10</ID>
</TSProperty>
```

4.15.2 Add additional POINTS sample

```
<TSProperty Type="POINTS">
     <ID>+15</ID>
</TSProperty>
```

4.15.3 Reduce number of POINTS sample

```
<TSProperty Type="POINTS">
     <ID>-20</ID>
</TSProperty>
```

4.15.4 Group tickets

The TSProperty POINTS is also used for Group Tickets. The POINTS value specifies the number of group members.

4.16 TSProperty DISPLAYMESSAGE

The TSProperty DISPLAYMESSAGE is used to display a text string at the reader's display when a ticket is checked and allowed to enter. This text overrides any other default display message configured within the HSH system. This property can be used to display different messages for each single ticket. The checkpoints can show the text at the 2 line display. The length and structure of the DISPLAYMESSAGE value depends on the used reader's capabilities. The character **\n** is used to separate the first and the second line. The TSProperty DISPLAYMESSAGE can only be used as part of the Whitelist (see WhitelistRecord samples).

4.16.1 DISPLAYMESSAGE sample

```
<TSProperty Type="DISPLAYMESSAGE">
     <ID>Welcome Mr.\n John Q. Public</ID>
</TSProperty>
```

4.17 TSProperty OPERATORMESSAGE

The TSProperty OPERATORMESSAGE is used to display a text string at the reader's operator display when a ticket was checked successfully. The operator display is an additional device mounted at the checkpoint. The OPERATORMESSAGE uses the same specifications as the DISPLAYMESSAGE.

4.17.1 OPERATORMESSAGE sample

Internal

```
<TSProperty Type="OPERATORMESSAGE">
     <ID>Ticket lost\n2018-12-31</ID>
</TSProperty>
```


Internal

4.18 TSProperty TIMETICKETPERIOD

This TSProperty is to assign an amount of time to a time ticket (e.g.: 3 hours, 30 minutes, etc.). The TSProperty TIMETICKETPERIOD informs HSH about the purchased amount of time. The unit for the TIMETICKETPERIOD is minutes. The TSProperty TIMETICKETPERIOD can only be used as part of the Whitelist (see WhitelistRecord samples).

If additional time value needs to be added to the existing time value, the property TIMETICKETPERIOD is used with a + sign in front of the ID.

If time value needs to be refunded (not used time period) actual time amount can be reduced by sending the property TIMETICKETPERIOD is used with a - sign in front of the ID.

Adding the **From** to the TIMETICKETPERIOD, the starting point for the time value deduction can be set. This makes it possible to start the time deduction at a certain time, e.g. when the ticket is printed.

4.18.1 TIMETICKETPERIOD sample

```
<TSProperty Type="TIMETICKETPERIOD">
     <ID>60</ID>
</TSProperty>
```

4.18.2 Add time sample

```
<TSProperty Type="TIMETICKETPERIOD">
     <ID>+60</ID>
</TSProperty>
```

4.18.3 Reduce time sample

```
<TSProperty Type="TIMETICKETPERIOD">
     <ID>-60</ID>
</TSProperty>
```

4.18.4 Set start time sample

```
<TSProperty Type="TIMETICKETPERIOD" From="2018-01-01T12:00:00">
     <ID>60</ID>
</TSProperty>
```

4.19 TSProperty TIMETICKETBEGIN

With TSProperty TIMETICKETBEGIN the start time for a time ticket can be defined. The used time is calculated using the **From** date as start time independent of the real entry date. With this feature it is possible to use time tickets even if no turnstile exists at the entrance because the start date is provided in the Whitelist.
TIMETICKETBEGIN can only be used as part of the Whitelist (see WhitelistRecord samples).

4.19.1 TIMETICKETBEGIN sample

```
<TSProperty Type="TIMETICKETBEGIN" From="2018-01-01T09:00:00" />
```


Internal

4.20 Use of TSProperties TIMETICKETBEGIN and TIMETICKETPERIOD

Both properties can be used in combination or alone.

4.20.1 Use of TSProperty TIMETICKETBEGIN alone

Using of XML attribute “From” is mandatory. “From” defines when a time ticket becomes valid.

```
<TSProperty Type="TIMETICKETBEGIN" From="2025-05-01T09:00:00" />
```

4.20.2 Use of TSProperty TIMETICKETPERIOD alone

Use of XML attribute “From” is mandatory and the mandatory Sub-element “ID” must contain a value for the time interval in minutes that will be added to the timestamp defined in “From”. “From” defines when a time ticket becomes valid.

```
<TSProperty Type="TIMETICKETPERIOD" From="2025-05-01T09:00:00">
     <ID>60</ID>
</TSProperty>
```

4.20.3 Use of TSProperties TIMETICKETBEGIN and TIMETICKETPERIOD in combination

For TSProperty TIMETICKETBEGIN use of XML attribute “From” is mandatory. “From” defines when a time ticket becomes valid. For TSProperty TIMETICKETPERIOD only the definition of time interval (XML element “ID”) is mandatory. The XML attribute “From” is not necessary in this case, because TIMETCIKETBEGIN already defines the beginning time.

```
<TSProperty Type="TIMETICKETBEGIN" From="2025-05-01T09:00:00" />
<TSProperty Type="TIMETICKETPERIOD">
     <ID>60</ID>
</TSProperty>
```

4.21 TSProperty PHOTO

The TSProperty PHOTO is used to import JPG images into the Handshake database. The photos are required for the photo control feature. The photo of the season pass holder is sent with the whitelist data. When the ticket is used the photo is displayed at the Handshake Monitor. Moreover, when the user inspects those transactions generated by using a ticket with a photo property, that photo is also shown in Handshake Monitor, Handshake Ticket Tracking, Handshake Photo Export and Handshake Reporting tools.

The TSProperty XML element contains the following child elements:

- Action specifies the type of action (insert, update, delete) that will be applied to the respective TSProperty.
- ID is the unique ID of the TSProperty generated by the ticket system. The ID identifies a photo, if the same ID is used again the photo is updated.
- Data is a Base64 binary representation of the JPEG image. The photo must meet the following requirements:
  - Ratio: 3:4
  - Dimensions: at least 300 x 400 pixel

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- Colors: up to 24 Bit
- File format: JPEG
- Photo must be passport photo like

The section 4.21.1 shows an example of a TSProperty of type PHOTO.

To fulfill the security requirements when handling with user related data, starting from Handshake version 09.01 photos are saved encrypted in the Handshake database. The Handshake tools that uses these photos are the only one able to decrypt them properly.

4.21.1 PHOTO sample

```
<TSProperty Type="PHOTO">
     <Action>I</Action>
     <ID>1</ID>
     <Data>/9j/4SXKRXhpZgAASUkqAAgAAAAJAA ...</Data>
</TSProperty >
```

4.22 TSProperties for Permission Inquiry Interface (PII)

When using Handshake.Logic together with a SKIDATA parking system Parking.Logic, tickets for the event can also be used at the parking columns. The PII interface is used to verify the validity of Handshake tickets in the parking area. For details of the Permission Inquiry Interface please see Permission-Inquiry-interface-3-3-en-tec.pdf.

Following properties can be used:

4.22.1 Mandatory: PARKINGTYPE or PARKINGARTICLECODE

Either the property PARKINGTYPE or the property PARKINGARTICLECODE must be given. If both properties are given, the PARKINGARTICLECODE has higher priority and overrides the PARKINGTYPE value.
The property PARKINGTYPE defines the type of the parking ticket. The PARKINGTYPE must have one of the following values:

- OneDayEntry
- OneDayExit
- MultipleExit
- SeasonPass
- LongRunPass

Note: The values are automatically assigned to a predefined article code in Parking.Logic. The articles are normally hidden and start with the character '~'.

The property PARKINGARTICLECODE (Handshake V5.1) allows specifying an arbitrary type of the parking ticket. The specified article code must exist in the parking system.

4.22.1.1 PARKINGTYPE sample

```
<TSProperty Type="PARKINGTYPE">
     <Action>I</Action>
     <ID>OneDayEntry</ID>
     <Name>One Day Entry Ticket</Name>
</TSProperty>
```

4.22.1.2 PARKINGARTICLECODE sample

```
<TSProperty Type="PARKINGARTICLECODE">
     <ID>4711</ID>
```


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```
</TSProperty>
```

4.22.2 TSProperty PARKINGVALIDITY

The PARKINGVALIDITY defines the valid time period for the parking ticket. The format is the same as the VALIDITY format.

4.22.2.1 PARKINGVALIDITY sample

```
<TSProperty Type="PARKINGVALIDITY" From="2018-01-01T00:00:00" To="2018-12-31T23:59:00" />
```

4.22.3 TSProperty PARKINGDISCOUNT

The PARKINGDISCOUNT (Handshake V5.1) defines a discount which is given to the parking fee. The discount can be specified either as monetary value or as percentage in the format “<discount type>;<discount value>”

- Possible discount-types: “Money”, “Percentage”
- Discount value: positive integer

4.22.3.1 PARKINGDISCOUNT sample

```
<TSProperty Type="PARKINGDISCOUNT">
     <ID>Money;2</ID>
     <!-- gives a discount of two monetary units (Euros, Dollars…) -->
</TSProperty>

<TSProperty Type="PARKINGDISCOUNT">
     <ID>Percentage;10</ID>
     <!-- gives a discount of 10% -->
</TSProperty>
```

4.22.4 TSProperty PARKINGTIMEWINDOWENTRY

The PARKINGTIMEWINDOWENTRY (Handshake V5.1) defines the allowed time range to enter the car park. It is specified as from-time (in hours and minutes) to to-time in the format “hh:mm;hh:mm”.

4.22.4.1 PARKINGTIMEWINDOWENTRY sample

```
<TSProperty Type="PARKINGWINDOWENTRY">
     <ID>16:00;21:25</ID>
     <!--car park entry is allowed between 16:00 and 21:25 -->
</TSProperty>
```

4.22.5 TSProperty PARKINGTIMEWINDOWPAID

The PARKINGTIMEWINDOWPAID (Handshake V5.1) defines a time range which is already paid and excluded from the parking fee calculation. It is specified as from-time to to-time in the format “hh:mm;hh:mm”.

4.22.5.1 PARKINGTIMEWINDOWPAID sample

```
<TSProperty Type="PARKINGWINDOWPAID">
     <ID>17:00;22:00</ID>
     <!--the time range between 17:00 and 22:00 is already paid -->
</TSProperty>
```

4.22.6 TSProperty PARKINGPAIDUNTIL

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The PARKINGPAIDUNTIL (Handshake V5.1) defines a timestamp – the parking fee is already paid until the specified date and time.

4.22.6.1 PARKINGPAIDUNTIL sample

```
<TSProperty Type="PARKINGPAIDUNTIL">
     <ID>2018-09-01T22:00</ID>
     <!-- is paid till defined date -->
</TSProperty>
```

4.22.7 TSProperty PARKINGTIMEVALUEPAID

The PARKINGTIMEVALUEPAID (Handshake V5.1) defines the number of minutes already paid. If the total parking time exceeds the specified number, only the remaining minutes are considered in the parking fee calculation.

4.22.7.1 PARKINGTIMEVALUEPAID sample

```
<TSProperty Type="PARKINGTIMEVALUEPAID">
     <ID>180</ID>
     <!-- 3 hours are paid -->
</TSProperty>
```

4.23 TSProperty for TEG

If the TEG is used (as an e-ticket gateway between the TS and HSH), then the TEG creates an additional TSProperty of type ET_STATUS and adds it to the TS properties of every permission of the whitelist record.

4.23.1 TSProperty ET_STATUS

The ET_STATUS provides information about the result of an e-ticket operation which was undertaken for a permission.
The TSProperty ID element contains the TEG specific error/warning number (see 4.4.3 for the according list) and the Name element contains the respective error/warning short description.
If multiple errors occur then the first error is reported.

4.23.1.1 ET_STATUS sample

```
<TSProperty Type="ET_STATUS">
     <Action>I</Action>
     <ID>22007</ID>
     <Name>MissingTicketDefinition</Name>
</TSProperty >
```


Internal

4.24 Permission element

The permission describes the sold ticket (e.g. 1 day adult ticket for friendly game including parking). One data carrier can have several permissions. Each permission requires a unique permission ID (UPID), we recommend starting with UPID 1, and to increment the value for each additional permission of a ticket. For season passes the event id can be used as UPID.

For tickets imported without UPId, the system will set the value 0 internally. Therefore it is strongly discouraged to use UPId = 0 in actual imports.

As only limited information can be stored in the barcode of the ticket, additional TSProperties can be provided as part of permission element, TSProperties stored within the barcode should also be sent. If properties on the ticket must be overwritten, the new TSProperties values must be provided within the Whitelist.

The elements TicketUsed, LastDate, Transaction and InquiryError of the Permission record are used with the Inquiry command.

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Ticket System – Handshake.Logic

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
UPID	Element	Yes	Yes	String	30	Unique Permission Id	1
Blocked	Attribute			String	10	No … valid not blocked permission Yes … blocked permission	No
MultiDC	Attribute			boolean	5	false … no multi data carrier permission (default) true… multi data carrier permission Note: works only for Whitelist import, Blacklist import, Permission inquiry, TimeTicketPeriod inquiry and TicketUsage inquiry	true
TSProperty	Element			Elem.		See TSProperty element	
TicketUsed	Element			boolean	1	Flag if ticket was already used 0 … not used 1 … used Used by Inquiry	0
LastDate	Element			dateTime	19	Date of last ticket usage (Optionally includes time zone information)	2018-12-31T01:33:00 (2018-12-31T01:33:00+04:00)
Transaction	Element			Elem.		Transaction data See Transaction element	
InquiryError	Element			Elem.		In case of error See InquiryError element	
Time	Element			Elem.		Contains information about time tickets; Used by Inquiry See Time element	

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4.24.1 Permission sample

Sample without MultiDC attribute:

```
<Permission>
     <UPID>1</UPID>
     <TSProperty Type="EVENT">
        <ID>3002</ID>
     </TSProperty>
</Permission>
```

Sample with MultiDC attribute:

```
<Permission MultiDC="true">
     <UPID>1</UPID>
     <TSProperty Type="EVENT">
        <ID>3002</ID>
     </TSProperty>
</Permission>
```

Note: The attribute “MultiDC” is only available for Whitelist import, Blacklist import and TicketUsage Inquiry.

4.25 WhitelistRecord element

When the Whitelist check is setup as mandatory in HSH, each ticket is only allowed for entry if TS has sent a WhitelistRecord. The TS transmits one Whitelist record per issued permission. The Whitelist is also used to carry additional information for single permissions (e.g. VALIDITY, ADDRELEASES).

HSH can be configured to check or not to check the existence of a Whitelist record as additional security feature. If tickets are coded with event related information, season tickets should have one permission for every single event within the Whitelist. Sending Whitelist information to HSH is optional but recommended.

Note: The Expire date defines if a WhitelistRecord is valid or not. Expired WhitelistRecords are ignored and treated as not existing.

The meaning of Expire date is special for the WhitelistRecord and is different to TSProperty and BlacklistRecord. In case of TSProperty and BlacklistRecord the Expire date is only relevant for the Handshake Archiver, a program that is used to backup and to delete old records.

Internal

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Expire	Attribute			dateTime	25	Date when property can be archived Expired records are ignored (Optionally includes time zone information)	2018-12-31T00:00:00 (2018-12-31T00:00:00+04:00)
Action	Element		Yes	string	1	I… insert D … delete U … update A … anonymize	I
RequestId	Element		Only if value of Action is A	i4	11	Unique request id Number provided by TS in order to link the request and the response	1234
UTID	Element	Yes	Yes	string	100	Unique ticket-ID	1234567890
Coding	Element	Yes	Yes	i4	11	Enumeration for type of ticket Code to identify a real ticket, see Coding definition table	257
Permission	Element	Yes		Elem.		Defines permission of ticket See Permission element	
ETicket	Attribute	No	No	boolean	5	Used in a TEG and MultiDC scenario to identify the DC becoming the e-ticket	true, false

4.25.1 Update WhitelistRecord

TSProperties that are used only once within the WhitelistRecord, can be updated. To be able to update a WhitelistRecord the attribute “Multiple identifier assignment” must not be active.

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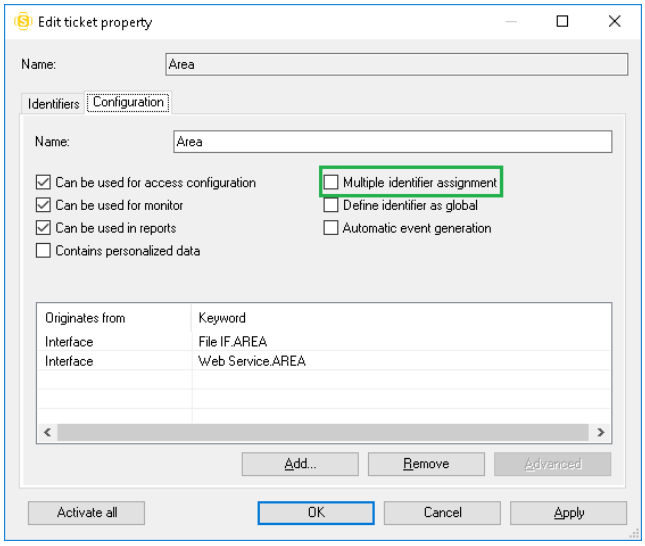
Ticket System – Handshake.Logic



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4.25.2 Update Expire date/invalidate existing permissions of tickets

Action U (Update) can be used to change the Expiry date of existing permissions. This feature can be used to get rid (invalidate) of existing permissions by setting the Expire date to a date in the past. Expired WhitelistRecords are ignored.

It is also possible to reset (set to null) the expired date of already imported permissions. In that case the Expire attribute is simply omitted.

4.25.2.1 Invalidate existing permissions of tickets sample

Invalidate all permissions of barcode ticket **30029659271.**

```
<WhitelistRecord Expire="2018-12-31T00:00:00">
     <Action>U</Action>
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
</WhitelistRecord>
```

With time zone information

```
<WhitelistRecord Expire="2018-12-31T00:00:00+04:00">
     <Action>U</Action>
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
</WhitelistRecord>
```

4.25.3 Anonymize personal properties of a WhitelistRecord

HSH provides the possibility to anonymize personal data on request. This can be done by transmitting a WhitelistRecord which contains a permission (multi data carrier permission) or UTID, Coding and UPID (single data carrier permission) which is associated with personal data of a customer. The Action **A** (Anonymize) can be used to anonymize personal data of existing permissions. Additionally, the ticket provider must send a unique RequestId which will be send back to ticket provider in response telegram when the data has been anonymized successfully.

To be able to anonymize personal properties of a WhitelistRecord, “GDPR” must be activated and the attribute “Contains personalized data” must be selected in the Handshake Explorer configuration.

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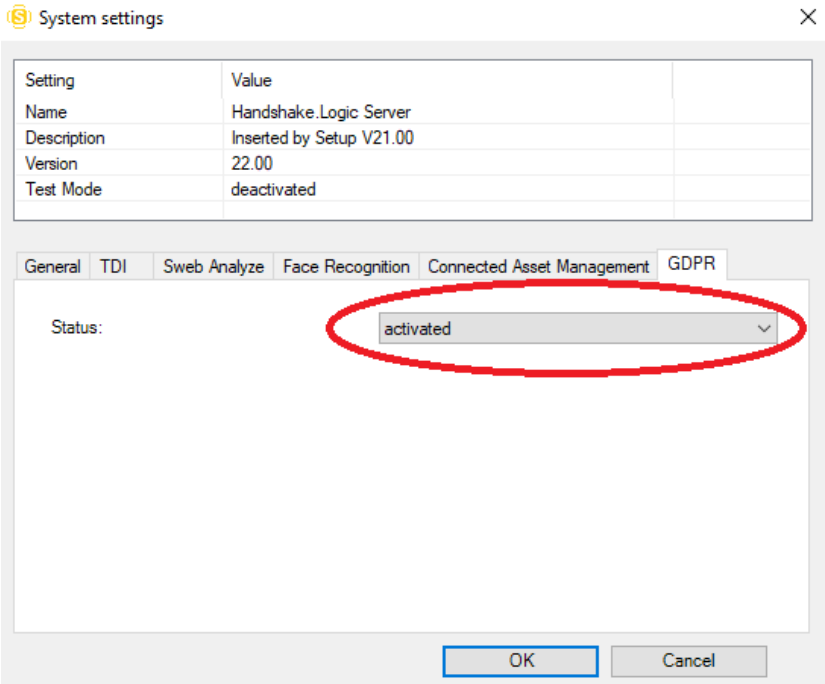

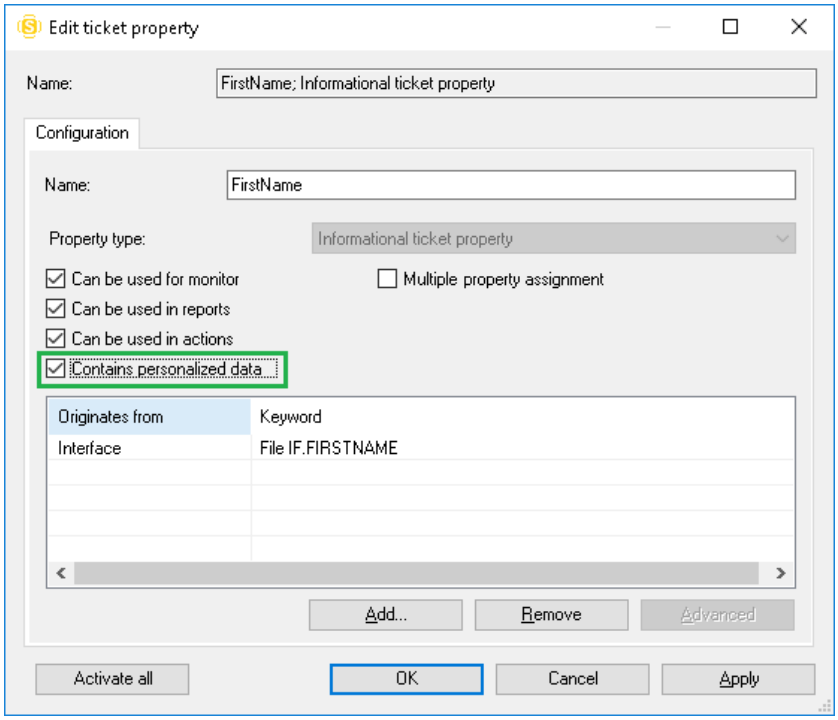

4.25.4 Result element for Anonymization request

The Result element is returned by a Whitelist import with action=A.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Result	Element		Yes	Elem.			
RequestId	Element	Yes	Yes	i4	11		1

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UPID	Element	Yes	Yes	String	30	Unique Permission Id	1
AffectedRows	Element		Yes	i4		Number of rows which have been changed for the given permission	1
SumAffectedRows	Element		Yes	i4		Sum of all rows which have been changed for the given permissions	3
Permission	Element			Elem.		Defines permission of ticket See Permission element	

4.25.5 WhitelistRecord samples

4.25.5.1 WhitelistRecord single permission ticket without additional TSProperties

```
<WhitelistRecord Expire="2018-12-31T00:00:00+04:00">
     <Action>I</Action>
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="false">
        <UPID>1</UPID>
        </Permission>
</WhitelistRecord>
```

4.25.5.2 WhitelistRecord update TSProperty EVENT

```
<WhitelistRecord Expire="2018-12-21T01:02:03+04:00">
     <Action>U</Action>
     <UTID>29649183693</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="false">
        <UPID>1</UPID>
        <TSProperty Type="EVENT">
           <ID>3010</ID>
        </TSProperty>
     </Permission>
</WhitelistRecord>
```

4.25.5.3 WhitelistRecord with several permissions (e.g. season pass)

```
<WhitelistRecord Expire="2018-12-31T00:00:00+04:00">
     <Action>I</Action>
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="false">
        <UPID>1</UPID>
        <TSProperty Type="EVENT">
           <ID>3001</ID>
        </TSProperty>
     </Permission>
     <Permission MultiDC="false">
        <UPID>2</UPID>
        <TSProperty Type="EVENT">
           <ID>3002</ID>
        </TSProperty>
     </Permission>
     <Permission MultiDC="false">
```

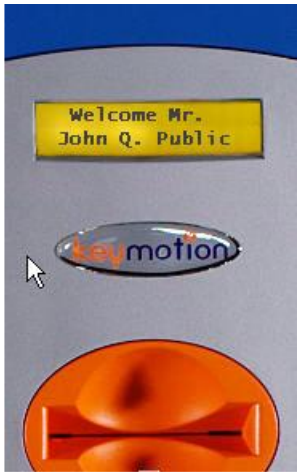

Internal

```
        <UPID>3</UPID>
        <TSProperty Type="EVENT">
           <ID>3003</ID>
        </TSProperty>
     </Permission>
</WhitelistRecord>
```

4.25.5.4 WhitelistRecord with additional TSProperties (not stored on ticket)

```
<WhitelistRecord Expire="2018-12-31T00:00:00+04:00">
     <Action>I</Action>
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="false">
        <UPID>1</UPID>
        <TSProperty Type="VALIDITY" From="2018-07-01T08:00:00" To="2018-07-01T19:00:00"/>
     </Permission>
</WhitelistRecord>
```

4.25.5.5 WhitelistRecord with DISPLAYMESSAGE




```
<WhitelistRecord Expire="2018-12-31T00:00:00+04:00">
     <Action>I</Action>
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="false">
        <UPID>1</UPID>
        <TSProperty Type="EVENT">
           <ID>1</ID>
        </TSProperty>
        <TSProperty Type="AREA">
           <ID>A</ID>
        </TSProperty>
        <TSProperty Type="DISPLAYMESSAGE">
           <ID>Welcome Mr.\n John Q. Public</ID>
        </TSProperty>
     </Permission>
</WhitelistRecord>
```

4.25.5.6 WhitelistRecord permissions with and without multi-data carrier

```
<WhitelistRecord Expire="2018-11-28T12:02:03+04:00">
     <Action>I</Action>
     <UTID>29649183693</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true">
        <UPID>1</UPID>
        <TSProperty Type="EVENT">
           <ID>3010</ID>
        </TSProperty>
     </Permission>
     <Permission MultiDC="false">
        <UPID>2</UPID>
        <TSProperty Type="EVENT">
           <ID>2999</ID>
        </TSProperty>
     </Permission>
</WhitelistRecord>
```


Internal

4.25.5.7 WhitelistRecord several data carriers – same multi-data carrier permission

```
<WhitelistRecord Expire="2018-11-28T12:02:03+04:00">
     <Action>I</Action>
     <UTID>29649183600</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true">
        <UPID>1</UPID>
        <TSProperty Type="EVENT">
           <ID>3000</ID>
        </TSProperty>
     </Permission>
</WhitelistRecord>
<WhitelistRecord Expire="2018-11-28T12:02:03+04:00">
     <Action>I</Action>
     <UTID>29649183601</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true">         <!-- same permission -->
        <UPID>1</UPID>
        <TSProperty Type="EVENT">
           <ID>3000</ID>
        </TSProperty>
     </Permission>
</WhitelistRecord>
```

4.25.5.8 WhitelistRecord several data carriers – same multi-data carrier permission, TSProperties only once

```
<WhitelistRecord Expire="2018-11-28T12:02:03+04:00">
     <Action>I</Action>
     <UTID>29649183600</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true">
        <UPID>1</UPID>
        <TSProperty Type="EVENT">        <!-- only once -->
           <ID>3000</ID>
        </TSProperty>
     </Permission>
</WhitelistRecord>
<WhitelistRecord Expire="2018-11-28T12:02:03+04:00">
     <Action>I</Action>
     <UTID>29649183601</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true">
        <UPID>1</UPID>
     </Permission>
</WhitelistRecord>
```

4.25.5.9 Delete a single permission data carrier

```
<WhitelistRecord>
     <Action>D</Action>
     <UTID>29649183600</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="false">
        <UPID>1</UPID>
     </Permission>
</WhitelistRecord>
```

4.25.5.10 Anonymize personal data of a single data carrier permission

Request from TS

```
<WhitelistRecord>
     <Action>A</Action>
     <RequestId>123</RequestId>
     <UTID>82719010418999</UTID>
     <Coding>257</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</WhitelistRecord>
```


Internal

Answer from HSH

```
<Result>
     <RequestId>123</RequestId>
     <Permission>
        <UPID>1</UPID>
        <AffectedRows>3</AffectedRows>
     </Permission>
     <SumAffectedRows>3</SumAffectedRows>
</Result>
```

4.25.5.11 Anonymize personal data of a single data carrier permission with several permissions (e.g. season pass)

Request from TS

```
<WhitelistRecord>
     <Action>A</Action>
     <RequestId>101</RequestId>
     <UTID>82719010418999</UTID>
     <Coding>257</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
     <Permission>
        <UPID>2</UPID>
     </Permission>
     <Permission>
        <UPID>3</UPID>
     </Permission>
</WhitelistRecord>
```

Answer from HSH

```
<Result>
     <RequestId>101</RequestId>
     <Permission>
        <UPID>1</UPID>
        <AffectedRows>1</AffectedRows>
     </Permission>
     <Permission>
        <UPID>2</UPID>
        <AffectedRows>1</AffectedRows>
     </Permission>
     <Permission>
        <UPID>3</UPID>
        <AffectedRows>1</AffectedRows>
     </Permission>
     <SumAffectedRows>3</SumAffectedRows>
</Result>
```

4.25.5.12 Anonymize personal data of a single data carrier permissions

Request from TS

```
<WhitelistRecord>
     <Action>A</Action>
     <RequestId>101</RequestId>
     <UTID>82719010418998</UTID>
     <Coding>257</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</WhitelistRecord>
<WhitelistRecord>
     <Action>A</Action>
     <RequestId>102</RequestId>
     <UTID>82719010418999</UTID>
     <Coding>257</Coding>
     <Permission>
        <UPID>2</UPID>
     </Permission>
</WhitelistRecord>
```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Answer from HSH

```
<Result>
     <RequestId>101</RequestId>
     <Permission>
        <UPID>1</UPID>
        <AffectedRows>1</AffectedRows>
     </Permission>
     <SumAffectedRows>1</SumAffectedRows>
</Result>
<Result>
     <RequestId>102</RequestId>
     <Permission>
        <UPID>2</UPID>
        <AffectedRows>1</AffectedRows>
     </Permission>
     <SumAffectedRows>1</SumAffectedRows>
</Result>
```

4.25.5.13 Anonymize personal data of a multi-data carrier permission

Request from TS

```
<WhitelistRecord Expire="2017-11-28T12:02:03+04:00">
     <Action>A</Action>
     <RequestId>124</RequestId>
     <Permission MultiDC="true">
        <UPID>28555</UPID>
     </Permission>
</WhitelistRecord>
```

Response:

```
<Result>
     <RequestId>124</RequestId>
     <Permission>
        <UPID>28555</UPID>
        <AffectedRows>1</AffectedRows>
     </Permission>
     <SumAffectedRows>1</SumAffectedRows>
</Result>
```

4.26 BlacklistRecord element

The tickets listed in the Blacklist are not allowed to enter. The content of the backlist is the same as the Whitelist extended with some useful information that may help the operator to handle blocked tickets at the entrance. The Whitelist information is needed to identify the blocked ticket or permission.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Expire	Attribute			dateTime	19	Date when property can be archived (Optionally includes time zone information)	2018-12-31T00:00:00 (2018-12-31T00:00:00 +4:00)
Action	Element		Yes	string	1	I… insert D … delete	I
Whitelist-Record	Element	Yes	Yes	Elem.		See WhitelistRecord element	

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Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Blocking Type	Element		Yes	i4	11	Enumeration: 0 … data carrier blocking 1 … permission blocking See Blocking types	1
From	Element			dateTime	19	Beginning of blocking (Optionally includes time zone information)	2018-10-12T12:00:00 (2018-10-12T12:00:00+04:00)
To	Element	Yes	Yes[1]	dateTime	19	End of blocking (Optionally includes time zone information)	2018-10-12T20:00:00 (2018-10-12T20:00:00+04:00)
Blocking Reason	Element		Yes	I4	11	Enumeration: 0 ... ticket blocked 1 … ticket was stolen 2 … ticket was lost 3 … ticket was cancelled 4 … ticket was renewed 5 … ticket was not paid 6 … ticket not extended 7 … ticket was exchanged 8 … substitute ticket 9 … terminated	5

[1] The To-Element is mandatory for a data carrier blocking (BlockingType = 0) only

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Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Display Message	Element			string	100	2 Lines with max.14 characters per line can be shown on the display of the reader ASx70. '\n' is used to separate the lines	Ticket not valid\n Contact office!
Comment	Element			string	100	Additional information shown to operator	He never pays in time
Operator Message	Element			string	100	Additional information shown to operator	He never pays in time

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4.26.1 BlacklistRecord samples

4.26.1.1 BlacklistRecord for ticket (data carrier), no MultiDC

All permissions of the ticket are indirectly blocked. Permissions (UPIDs) would be ignored if added to request.
Required Parameters: UTID, Coding, BlockingType = TicketBlocking, ToDate, BlockingReason

```
<BlacklistRecord>
        <Action>I</Action>
        <WhitelistRecord>
                <Action>I</Action>
                <UTID>12345</UTID>
                <Coding>260</Coding>
        </WhitelistRecord>
        <BlockingType>0</BlockingType>
        <BlockingReason>2</BlockingReason>
        <To>2021-12-31T00:00:00</To>
 </BlacklistRecord>
```

4.26.1.2 BlacklistRecord several permissions, no MultiDC

The UPID(s), given for the ticket, are explicitly blocked. The permissions (WhitelistRecord) must be imported previously.
Required Parameters: UTID, Coding, UPID(s), BlockingType = PermissionBlocking, BlockingReason

```
<BlacklistRecord>
        <Action>I</Action>
        <WhitelistRecord>
                <Action>I</Action>
                <UTID>12345</UTID>
                <Coding>260</Coding>
                <Permission MultiDC="False">
                        <UPID>1</UPID>
                </Permission>
                <Permission MultiDC="False">
                        <UPID>2</UPID>
                </Permission>
                <Permission MultiDC="False">
                        <UPID>3</UPID>
                </Permission>
        </WhitelistRecord>
        <BlockingType>1</BlockingType>
        <BlockingReason>2</BlockingReason>
        <To>2021-12-31T00:00:00</To>
</BlacklistRecord>
```

If a ticket with a non multi-data carrier permission is blacklisted and the permission does not exist in the Handshake system, then the permission and the permission block are created.

4.26.1.3 BlacklistRecord several permissions with MultiDC

The UPID(s) are blocked, related ticket(s) are indirectly blocked.
Required Parameters: UPID(s), BlockingType = PermissionBlocking, BlockingReason

```
<BlacklistRecord>
        <Action>I</Action>
        <WhitelistRecord>
                <Action>I</Action>
                <UTID/>
                <Coding>0</Coding>
                <Permission MultiDC="True">
```


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```
                        <UPID>Unique1</UPID>
                </Permission>
                <Permission MultiDC="True">
                         <UPID>Unique2</UPID>
                </Permission>
                <Permission MultiDC="True">
                         <UPID>Unique3</UPID>
                </Permission>
        </WhitelistRecord>
        <BlockingType>1</BlockingType>
        <BlockingReason>2</BlockingReason>
        <To>2021-12-31T00:00:00</To>
</BlacklistRecord>
```

The precondition in this case is an existing multi-data carrier permission in the Handshake system otherwise an error “non-existing permission”.

4.26.1.4 Delete a BlacklistRecord (data carrier), no MultiDC

All permissions of the ticket are indirectly unblocked

```
<BlacklistRecord>
       <Action>D</Action>
       <WhitelistRecord>
              <Action>I</Action>
              <UTID>12345</UTID>
              <Coding>260</Coding>
       </WhitelistRecord>
       <BlockingType>0</BlockingType>
       <BlockingReason>4</BlockingReason>
       <To>2021-12-31T00:00:00</To>
</BlacklistRecord>
```

The ticket “12345” is not blacklisted. The access with this ticket is allowed if it has a valid (not used) permission.

4.26.1.5 Delete a BlacklistRecord for specific permissions of a ticket, no MultiDC

The UPID(s), given for the ticket, are explicitly unblocked.
Required Parameters: UTID, Coding, UPID(s), BlockingType = PermissionBlocking, BlockingReason

```
<BlacklistRecord>
        <Action>D</Action>
        <WhitelistRecord>
                <Action>I</Action>
                <UTID>12345</UTID>
                <Coding>260</Coding>
                <Permission MultiDC="False">
                        <UPID>1</UPID>
                </Permission>
                <Permission MultiDC="False">
                        <UPID>2</UPID>
                </Permission>
                <Permission MultiDC="False">
                        <UPID>3</UPID>
                </Permission>
        </WhitelistRecord>
        <BlockingType>1</BlockingType>
        <BlockingReason>4</BlockingReason>
        <To>0001-01-01T00:00:00</To>
</BlacklistRecord>
```


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4.26.1.6 Delete a BlacklistRecord for a MultiDC permission

The UPID(s) are unblocked, related ticket(s) are indirectly unblocked.

Required Parameters: UPID(s), BlockingType = PermissionBlocking, BlockingReason

```
<BlacklistRecord>
        <Action>D</Action>
        <WhitelistRecord>
                <Action>I</Action>
                <UTID/>
                <Coding>0</Coding>
                <Permission MultiDC="True">
                        <UPID>Unique1</UPID>
                </Permission>
                <Permission MultiDC="True">
                         <UPID>Unique2</UPID>
                </Permission>
                <Permission MultiDC="True">
                         <UPID>Unique3</UPID>
                </Permission>
        </WhitelistRecord>
        <BlockingType>1</BlockingType>
        <BlockingReason>4</BlockingReason>
        <To>0001-01-01T00:00:00</To>
</BlacklistRecord>
```

4.26.2 Blocking types

The Blacklist can be filled with two different blocking types “Data Carrier Blocking” and “Permission Blocking”.

4.26.2.1 Permission blocking

Using “permission blocking” blocks only a certain permission of the ticket. The fields „From” and „To” may be left empty.
E.g.: A season ticket is blocked for an UEFA cup game, but it is still valid for other games.

4.26.2.2 Data Carrier blocking

The whole data carrier (keycard, swatch access, etc.) is blocked. This means that all associated permissions are blocked as well. Using this Blacklist type the column „To” must contain a valid date.
E.g.: A season ticket was lost.

4.27 Inquiry element

The Inquiry Element enables the TS to request data from HSH. This can be used to get information about a ticket usage in HSH.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Type	Attribute	Yes	Yes	string	100	Fix keywords: TicketUsage Transactions Permissions TimeTicketPeriod Points Days	TicketUsage

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Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
						Counter Imports Information TicketUsage: tells if a ticket was already used. Transactions: returns all transactions of a ticket	
Option	Attribute			i4	11	Used to suppress not required information; if this attribute is not used all described information is sent back to the TS See Option	5
Status	Attribute			String	20	All … all permissions Blocked … blocked permissions Valid … non blocked permissions	All
From	Attribute			dateTime	19	Beginning of inquired time period (Optionally includes time zone information)	2018-07-01T08:00:00 (2018-07-01T08:00:00 +04:00)
To	Attribute			dateTime	19	End of inquired time period (Optionally includes time zone information)	2018-07-31T20:00:00 (2018-07-31T20:00:00 +04:00)
Timestamp	Attribute			dateTime	19	Date and time when Inquiry telegram was created (Optionally includes time zone information)	2018-01-31T09:41:23 (2018-01-31T09:41:23 +04:00)

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Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Last-TransactionId	Attribute			i4	11	Id of last received transaction	956
Max-Transactions	Attribute			i4	11	Maximum number of transactions to return (default 10000)	100
MaxTransactionId	Attribute			I4	11	Maximum Id over all received transactions	1056
NotComplete	Attribute			i4	11	If value is 1 not all existing tickets were returned by Inquiry due to limitation of ticket number returned (default 10000)	1
UTID	Element			string	100	Used to identify ticket	123456
Coding	Element			i4	11	Used to identify ticket	257
Permission	Element			Elem.		Used to request special permissions. Using UPId=0 In this element will be treated as no UPId was given See Permission element	1
Ticket	Element			Elem.		Identifies a ticket See Ticket element	
Counter	Element			Elem.		All defined counters of the access system See Counter element	
Totals	Element			Elem.		See Totals element	
TSProperty	Element			Elem.		See TSProperty element	

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Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
GroupBy	Element			Elem.		Used with Inquiry of type Imports to group the result by values of the TSProperty specified in the GroupBy. See Imports.	

Hint: For performance reasons it is not allowed to request the full information for the inquiry types “TicketUsage” and “Transactions” by sending an Inquiry using just the „Type” attribute. At least a duration (From - To), UTID or LastTransactionId must be used within the Inquiry.

4.27.1 Type

The Type attribute selects the Inquiry type. Two main data categories can be inquired:

- Configuration based data
- Transactions based data

4.27.1.1 TicketUsage

With the TicketUsage Inquiry the TS requests information if a ticket was used or not. This telegram can be use at the POS to ensure that a ticket was not used before it is canceled.

4.27.1.2 Transactions

The Transaction inquires all transactions stored in the HSH database. This information can be used within the TS to create access reports. Within the Inquiry only transactions of tickets issued by the TS are returned by default.

To inquire transactions from other TS a configuration within HSH (using the HSH Explorer) needs to be made. To get all transactions Option Bit 10 must be used.

Note: The number of transactions returned is limited (by default 10000 transactions). If not all tickets (transactions) were returned by the Inquiry, the attribute NotComplete=”1” is added to the Inquiry element.

4.27.1.3 Permissions

This Inquiry returns the number of permissions (UPIDs) and all related details (TSProperties) of a data carrier (UTID). With the attribute Status you can define which records you are interested in.
The Blocked attribute of element Ticket is set to Yes only when the data carrier is blocked, regardless if all permissions of the ticket are blocked.
The Attribute Blocked of element Permission indicates if a returned permission is blocked (Blocked=”Yes”) or if the permission is valid (Blocked=”No”).

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- Status="All"
  All Permissions are returned by the Inquiry blocked and valid.
- Status="Valid"
  Only valid Permissions are returned by the Inquiry.
- Status="Blocked"
  Only blocked Permissions are returned by the Inquiry.

4.27.1.4 TimeTicketPeriod

This Inquiry is used in connection with time tickets. When the ticket's usage exceeds the purchased time period, the turnstile blocks the ticket when used for exit. The customer is informed that the time limit is exceeded. To calculate the overtime and the additional payment, the TS use the TimeTicketPeriod Inquiry. The result of this Inquiry is the purchased and the used amount of time.

4.27.1.5 Days

The Day Inquiry returns the number of days assigned to a day ticket, the number of purchased and used days.

4.27.1.6 Points

The Points Inquiry returns the number of points assigned to a point ticket, the number of purchased and used points.

4.27.1.7 Counter

The Counter Inquiry returns the number of passages of permissions (tickets) at the different locations (Venue, Area, Event, Gate, Checkpoint) as well as the configured capacity limits.

4.27.1.8 Imports

The Imports Inquiry returns the number of records received for verification purposes. A period and one or more TSProperties of interest can be defined. By default, all records received at the current day are taken into consideration.

The sections 0 to 0 show a few examples of using inquiry of type Imports.

The Imports Inquiry can specify an optional XML element "GroupBy" with a TSProperty attribute, e.g. `<GroupBy TSProperty="PERSONCATEGORY"/>`. In this case, the Imports inquiry returns the number of whitelist and blacklist records grouped by the values of the specified TSProperty. Section 0 shows an example of the usage of an Inquiry Imports with GroupBy.

4.27.1.9 Information

The information inquiry returns all stored information (personalized data) included in Handshake (permissions and transactions) and the third party to which the personalized data has been forwarded to. If the personalized data has been anonymized already, then Handshake cannot return any data on a request for information. In that case the response telegram is empty.

4.27.2 LastTransactionId

The attribute TransactionId is an HSH internal, unique ongoing number. Each ticket usage (ticket transaction) registered by HSH is connected to a unique TransactionId and can therefore be clearly identified. This TransactionId is also used for HSH internal purpose and therefore may result in gaps of IDs when inquiring the data.
Using the attribute LastTransactionId it is possible to inquire all new recorded transactions since the last Inquiry. If more than the maximum number of transactions (default 10000

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transactions) were used in the meantime, only the allowed maximum number of transactions is returned by the Inquiry. This is indicated with the attribute NotComplete set to 1 in the Inquiry response. New Inquiry telegrams must be sent to Handshake till all transactions - indicated by the missing attribute NotComplete or attribute NotComplete set to 0 - are returned.
When the inquiry returns no transactions, the values LastTransactionId and MaxTransactionId in the result will be set to the value of LastTransactionId of the request.

4.27.3 MaxTransactions

The MaxTransactions attribute can be used to limit the maximum number of transactions in an Inquiry if not all transactions returned are needed. The upper limit for the value of this attribute is HSH's internal transaction limit (default 10000).
The attribute can be used together with the Option Bit 17 (see chapter 4.27.4 Option) to inquire the last n transactions that fulfill the specified criteria.

4.27.4 Option

The Option attribute helps the TS to control which data is returned by HSH on the Inquiry request.

Bit	Description
0	Suppress RawData
1	Suppress Checkpoint information
2	Suppress Gate information
3	Suppress Area information
4	Suppress Venue information
5	Suppress TSProperty information
6	Suppress detailed permission information, return only number of permissions
7	New XML layout (will become standard);
8	Request all transactions include erroneous records (default only successful transactions)
9	Return also user defined counters (by default only system counters are returned)
10	Return also transactions of other Ticketing Systems (default only own transactions)
11	Return also ResultCodes of extended Checks (e.g. biometric checks)
12	Return BelongsTo date for variable day change
13	Return Timestamp date
14	Return permission type, current and used values
15	Return TSProperties instead of Handshake Properties
16	Return HSPropertyTypeId for Photo import
17	Return newest transactions first (by default oldest first)
18	Return all used conditions for each transaction

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Bit	Description
19	Return also Name of TSProperty Identifiers

4.27.4.1 New XML layout

If the new XML layout (Option=”128”) is used, the element Ticket includes all permissions of a single ticket. Using the original layout there is no difference between single tickets. Although the XML document is valid without the Ticket element, BizTalk does not support this structure. For compatibility reasons we still accept the old structure but recommend using the new layout. This new layout style is the standard layout.

4.27.4.2 Return TSProperty names

The standard inquiry returns the property names as configured in Handshake, these might differ from the TSProperty names as used in the interface. If Option Bit 15 (=32768) is set the property names match with the TSProperties used by the WhitelistRecord. Use Option Bit 19 (=524288) in addition to get also the Name of the TSProperty Identifiers.

Note: To use option bit 15 one interface keyword must be assigned to one property (1:1) within the HSH configuration, assignment of several interface keywords to 1 property (n:1) is not supported and you may get the keyword of a other interface.

4.28 Ticket element

The Ticket Element is the answer to the transaction inquiry. The Ticket Element includes all transactions of one single ticket.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Blocked	Attribute			string	10	No … valid not blocked permission Yes … blocked permission	No
UTID	Element	Yes	Yes	string	100	Unique ticket-ID	1234567890
Coding	Element	Yes	Yes	i4	11	Enumeration for type of ticket Code to identify a real ticket See Coding definition table	257
RawData	Element			Elem.		Information about the native ticket data stored on the data carrier (e.g. type of Coding, barcode data)	<DC><L T="0"><D T="1">407788825300000

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Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
						See RawData element	016</D></L></DC>
Permission	Element			Elem.		Defines permission of ticket See Permission element	
NumberOfUPID	Element			Elem.		Number of existing permissions for one ticket; Elem. is used by Inquiry	1

4.28.1 RawData element

The RawData element contains detailed information about the original data stored on the ticket. The information is stored as XML string, the layout of the XML string differs depending on the used coding technology, barcode or RF.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
DC	Element	Yes	Yes	Elem.	4000	Root element, abbreviation for Data Carrier See The DC (Data Carrier) element	

4.28.1.1 The DC (Data Carrier) element

The DC Element contains the information about the data carrier itself.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
CI	Attribute		Yes	i4	11	Chip Id for RF data carrier; used to distinguish different data carrier like keycard, Mifare, Felica …	8
SN	Attribute		Yes	i8	11	Serial number of RF data carrier	14466735
L	Element	Yes	Yes	Elem.		Abbreviation for Layer See L(ayer) element	

4.28.1.2 L(ayer) element

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The L element contains the original data of each layer.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
T	Attribute		Yes	i4	11	Type 0 … Barcode 1 … RF 2 … Magstripe	0
D	Element		Yes	Elem.		Contains native data See D(ata) element	

4.28.1.3 D(ata) element

The D element contains the barcode or magnetic stripe data or the data stored on the RF data carrier.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
D	Element		Yes	Elem.		Data	2510671567
T	Attribute		Yes	i4	11	Type same as Coding	6
X	Attribute			i4	11	X-position of barcode	120
Y	Attribute			i4	11	Y-position of barcode	100
E	Attribute			i4	11	Error code if reading of ticket fails	3
L	Attribute		Yes	i4	11	Number of data level	1
A	Attribute		Yes	i4	11	Application Id of RF data level	30
C	Attribute		Yes	i4	11	Company Id of RF data level	1
E	Attribute		Yes	date	10	Expiry date for RF data level	2018-12-31
I	Attribute			boolean	11	Invalidate 0 … Valid data level 1 … Invalid data level E.g. if a ticket is cancelled the data level is invalidated; Invalidated data levels can be overridden with a new permission	0

4.28.2 DC (data carrier) sample

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4.28.2.1 Barcode sample

```
<DC>
     <L T="0">
        <D T="1">4310041238339998</D>
     </L>
</DC>
```

4.28.2.2 Keycard sample

```
<DC CI="26" SN="16147115512906773038">
     <L T="1">
        <D L="2" A="30" C="1" E="2018-12-31">57F083489808000B18002048000000000000000000000000</D>
     </L>
</DC>
```

4.29 Transaction element

The Transaction element shows statistical data about HSH entries to the TS. This information is not only the counter for entries and exits, it provides detailed information about when and where the ticket was used. It also provides information if the ticket was checked online and if the transaction was successful yes or no.

Name	Attribute/ Element	Key	Mand.	Data Type	Max. Length	Remark	Example
TransactionId	Attribute	Yes	Yes	i4	11	Unique id for transaction	1
Date	Attribute		Yes	dateTime	19	Date and time of transaction	2018-01-13T08:31:54
BelongsTo	Attribute			dateTime	19	Logical date and time of transaction if day change is different to the standard 00:00	2018-01-13T08:31:00
Status	Attribute			i4	11	See Status	0
Validated	Attribute			i4	11	0 … validated online 1 … validated offline	0
Releases	Attribute			i4	11	Number of turnstile releases	1
PermissionType	Attribute			string		Permission type related to used ticket (permission) PermissionTypes: DayTicket SeasonPass PointTicket	Point Ticket

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Name	Attribute/ Element	Key	Mand.	Data Type	Max. Length	Remark	Example
						TimeTicket GroupTicket	
CurrentValue	Attribute			i4	11	Remaining value after passage (not for SeasonPass)	2
UsedValue	Attribute			i4	11	Value required for the passage (not for SeasonPass)	1
Venue	Element			Elem.		Venue information See Venue element	
Area	Element			Elem.		Area information See Area element	
Gate	Element			Elem.		Gate information See Gate element	
Checkpoint	Element			Elem.		Checkpoint information See Checkpoint element	

4.29.1 Transaction with PermissionType and value information sample

```
<Transaction TransactionId="2" Date="2018-01-25T07:43:09" Status="0" Validated="0" Releases="1"
             PermissionType="PointTicket" CurrentValue="9" UsedValue="1"/>
```

4.29.2 Status

The Status shows information about the successful or not successful transaction. The following table shows the possible status numbers and their meaning.

Stat.	Name	Description
0	No Error	The transaction succeeded without any problem.
31	Ticket not readable	Ticket data can't be read
32	Ticket data unknown	Barcode or RF data unknown
33	Ticket type unknown	There was an attempt to enter with a ticket which is not known by HSH
34	Internal System error / Ticket not allowed	Internal system error of access system

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Stat.	Name	Description
35	Ticket not allowed	
36	Extended check failed	An extended check is configured for the ticket and this check fails
37	Ticket manipulated	Ticket was manipulated
38	Ticket write error	Write/update ticket data failed (RFID cards)
51	Control card accepted	A control card was used
52	Control card unknown	The used control card was not configured in HSH
53	Missing or invalid print layout	Required print layout is missing
54	Ticket inserted wrong into reader	
61	Manual Release	
62	Passages while free passage	Checkpoint is in mode free passage (turnstile opens automatically without ticket), but ticket was used to pass
63	Forced passage	Turnstile was opened with force
64	Passage with additional release	Turnstile was opened because of an additional release
65	Passage with group ticket	Group ticket passage
66	Permanent open passage	Turnstile was passed during permanent open mode
103	Ticket check: access not allowed	Access not allowed because of missing permission in HSH
104	Ticket check: Access denied	Access not allowed because of a configured denial
105	Ticket check: day ticket, long term ticket valueless	Ticket is not valid any longer
106	Ticket check: no WhitelistRecord	WhitelistRecord is missing
107	Checkpoint off-line / Entry not allowed	Checkpoint is in offline mode
108	Ticket check: permission type unknown	No matching permission defined
109	Ticket check: validity check failed	Ticket has expired
110	Venue: entries exceeded	Number of defined Venue entries exceeded
111	Venue: entry waiting period is not yet up	The configured waiting period for reentry is not yet up
112	Venue: double use waiting period to enter is not yet up	The configured double usage waiting period for reentry is not yet up
113	Venue: exit waiting period is not yet up	The configured waiting period for re-exit is not yet up

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Stat.	Name	Description
114	Venue: double use waiting period to exit is not yet up	The configured double usage waiting period for re-exit is not yet up
115	Area: entries exceeded	Number of defined Area entries exceeded
116	Area: Entry waiting period is not yet up	The configured waiting period for reentry is not yet up
117	Area: double use waiting period to enter is not yet up	The configured double usage waiting period for reentry is not yet up
118	Area: exit waiting period is not yet up	The configured waiting period for re-exit is not yet up
119	Area: double use waiting period to exit is not yet up	The configured double usage waiting period for re-exit is not yet up
120	Event: entries exceeded	Number of defined Event entries exceeded
121	Event: entry waiting period is not yet up	The configured waiting period for reentry is not yet up
122	Event: double use waiting period to enter is not yet up	The configured double usage waiting period for reentry is not yet up
123	Event: exit waiting period is not yet up	The configured waiting period for re-exit is not yet up
124	Event: double use waiting period to exit is not yet up	The configured double usage waiting period for re-exit is not yet up
125	Venue: today's maximum of entries exceeded	Number of defined Venue entries for today exceeded
126	Area: today's maximum of entries exceeded	Number of defined Area entries for today exceeded
127	Event: today's maximum of entries exceeded	Number of defined Event entries for today exceeded
128	Ticket check: data carrier blocked	Ticket is blocked using data carrier blocking
129	Ticket check: permission blocked	Permission is blocked
130	Ticket check: condition failed	Access not allowed due to a configured condition
131	Ticket check: insufficient point credit	Point/Value ticket has insufficient number of points/values
132	Ticket check: point ticket valueless	Point/Value ticket has no points/values left
133	Entry without prior leaving	Exit is missing (checkin/checkout configuration)
134	Rollback of former release	
135	TimeTicket: time exceeded	
136	No TimeTicket	
137	TimeTicket: checkin/checkout ticket already left	

Internal

Stat.	Name	Description
138	TimeTicket: checkin/checkout ticket already entered	
139	TimeTicket: checkin/checkout ticket not entered yet	
140	Informative check OK	
141	Exit not possible	
142	Ticket already canceled	
145	Permission expired	Existing permission not valid anymore (Expire date in the past)
150	Counter: reached counter limit	
175	Control card: manual release	
200	Registration error, ticket not valid	Ticket not valid due to internal error
201	Out of Licenses	License problem e.g. number of used checkpoints

This list may be extended as new requirements will come up. Please check the partner web for new releases of this document.

All successful transactions are returned by the Inquiry by default (all transactions where releases greater than 0). Within the HSH Explorer a configuration can be made to define which unsuccessful transactions are exported.

4.29.3 Validated

The attribute Validated specifies if a transaction was done in online or offline mode of the checkpoint. If the value for Validated equals “0” the ticket was validated during online mode of the checkpoint, this means that the connection to the HSH server was ok and all information stored in the Handshake database was available when the ticket was checked. If the connection to the Handshake server is interrupted the ticket is checked by the reader software and the attribute Validated is set to value “1”. During the offline mode period the information stored on the ticket is used by the validation process.

4.29.4 Releases

The attribute Releases contains the number of turnstile releases (passages). To calculate the number of Entries and Exits the value of Releases must be used.

4.30 Venue element

A venue consists of one or more areas which are entered through gates and is an own unit. Within one HSH system several venues can be managed. The single venues cannot be connected. In most cases one venue is used.

Internal

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Name	Attribute	Yes	Yes	string	30	Name of venue	Arena1
Entry	Attribute		Yes	boolean	1	Specifies if entry otherwise exit	0
Exit	Attribute		Yes	boolean	1	Specifies if exit otherwise entry	1
FE	Attribute			boolean	1	Specifies if ticket was used the very first time (First Entry)	0
FDE	Attribute			boolean	1	Specifies if ticket was used the first time at the current day (First Day Entry)	0

4.31 Area element

An area consists of one or more gates and describes a zone within a venue.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Name	Attribute	Yes	Yes	string	30	Name of area	Area A1
Type	Attribute		Yes	boolean	1	0… Left Area 1… Entered Area	0
Entry	Attribute		Yes	boolean	1	Specifies if entry otherwise exit	0
Exit	Attribute		Yes	boolean	1	Specifies if exit otherwise entry	1
FE	Attribute			boolean	1	Specifies if ticket was used the first time	0
FDE	Attribute			boolean	1	Specifies if ticket was used the first time at the current day	0

4.32 Gate element

A gate consists of one or more checkpoints and must be passed to enter or leave a certain area. The gate marks the border of area(s).

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example

Internal

Name	Attribute	Yes	string	30	Name of gate	Reentry Gate

4.33 Checkpoint element

A checkpoint is the physical reader (e.g. Flex.Gate, Vario.Gate, Mobile.Gate, etc.) and barrier (e.g. SKIDATA turnstile, no barrier …) and executes the access control.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Name	Attribute		Yes	string	30	Name of checkpoint	Checkpoint 1

4.34 Time element

The Time element contains time information of time tickets. This Inquiry is used to get information about the purchased and consumed time. With this information a POS system should be able to perform the calculation of extra fees.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Purchased	Attribute		Yes	i4	11	Purchased time in minutes	60
Used	Attribute		Yes	i4	11	Used time in minutes	30

4.35 Counter element

The Counter element inquires all defined counters from HSH showing the current number of visitors as well as the defined capacity limits of the corresponding locations. This information can be used to control the ticket sales based on the current number of people inside the venue.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Id	Attribute		Yes	i4	11	Unique Id of counter	1
Name	Attribute		Yes	string	30	name of counter	Venue counter
Type	Attribute		Yes	i4	11	0 ... System counter 1 ... User defined counter 2 … Result code counter	0
Location	Element		Yes	Elem.		See Location element	
Entries	Element		Yes	i4	11	Number of entries	120
Exits	Element		Yes	i4	11	Number of exits	12

Internal

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Current	Element		Yes	i4	11	the number of people currently inside a venue, layout, area, or event, or number of passing at checkpoint or gate (V6.0)	12
UpperLimit	Element			i4	11	Maximum number of persons	1000
LowerLimit	Element			i4	11	Limit when lock is released and persons may enter again	900
Sold	Element			i4	11	Number of sold tickets (V6.0)	212
FirstEntries	Element			i4	11	Number of first entries (V6.0)	115
FirstEntries Day	Element			i4	11	Number of first entries on the current day (V6.0)	75
Locked	Element		Yes	i4	11	0 … open 1 … locked (currently no passage possible at Handshake™)	0

4.35.1 Counter sample

Request from TS

```
<Inquiry Type="Counter" />
```

Answer from HSH

```
<Inquiry Type="Counter">
     <Counter Id ="1" Name="Venue1" Type="0" >
        <Location Id="1" Name="Venue1" Type="Venue"/>
        <Entries>121</Entries>
        <Exits>20</Exits>
        <Current>101</Current>
        <UpperLimit>1000</UpperLimit>
        <LowerLimit>950</LowerLimit>
        <FirstEntries>115</FirstEntries>
        <FirstEntriesDay>75</FirstEntriesDay>
        <Locked>0</Locked>
     </Counter>
     <Counter Id ="2" Name="Area A" Type="0" >
        <Location Id="77" Name="Area A" Type="Area"/>
        <Entries>121</Entries>
        <Exits>20</Exits>
        <Current>101</Current>
        <UpperLimit>1000</UpperLimit>
        <LowerLimit>950</LowerLimit>
        <Locked>0</Locked>
     </Counter>
```


Internal

```
    <Counter Id ="3" Name="Gate A" Type="0" >
       <Location Id="78" Name="Gate A" Type="Gate"/>
       <Entries>121</Entries>
       <Exits>20</Exits>
       <Current>101</Current>
       <UpperLimit>1000</UpperLimit>
       <LowerLimit>950</LowerLimit>
       <Locked>0</Locked>
    </Counter>
    <Counter Id ="4" Name="CP 1" Type="0" >
       <Location Id="79" Name="CP 1" Type="Checkpoint"/>
       <Entries>100</Entries>
       <Exits>10</Exits>
       <Current>110</Current>
       <UpperLimit>1000</UpperLimit>
       <LowerLimit>950</LowerLimit>
       <Locked>0</Locked>
    </Counter>
    <Counter Id ="163" Name="Whitelist counter adults" Type="1" >
       <Location Id="80" Name="Venue1" Type="Venue"/>
       <Entries>55</Entries>
       <Exits>4</Exits>
       <Current>51</Current>
       <UpperLimit>1000</UpperLimit>
       <LowerLimit>950</LowerLimit>
       <Sold>212</Sold>
       <Locked>0</Locked>
    </Counter>
</Inquiry>
```

4.35.2 Counter sample with location

Available since Handshake V09.01. Location id and type are optional parameters and can be used to filter for specific counters.

Valid values for location type are: *Venue, Area, Gate, Checkpoint.*

Request from TS

```
<Inquiry Type="Counter">
    <Location Id="80" Type="Venue"/>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Counter">
    <Counter Id ="163" Name="Whitelist counter adults" Type="1" >
       <Location Id="80" Name="Venue1" Type="Venue"/>
       <Entries>55</Entries>
       <Exits>4</Exits>
       <Current>51</Current>
       <UpperLimit>1000</UpperLimit>
       <LowerLimit>950</LowerLimit>
       <Sold>212</Sold>
       <Locked>0</Locked>
    </Counter>
</Inquiry>
```

4.36 Location element

The Location Element contains the location details of a Counter. To get the total amount of visitors the venue counter (Counter Type=”0” and Location Type=”Venue”) can be used, if more than one venue exists the counters have to be added up.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example

Internal

Id	Attribute		Yes	i4	11	Unique Id of Location	1
Name	Attribute		Yes	string	30	Name of Location	Venue counter
Type	Attribute		Yes	string	30	Venue Area Event (if configured) Gate Checkpoint	Venue

4.37 Totals element

The Totals element is returned by the Imports Inquiry and contains the number of Whitelist Records and BlacklistRecords. BlacklistRecords are separated in BlockingType 0 for data carrier blocking and BlockingType 1 for permission blocking.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Totals	Element		Yes	Elem.			
WhitelistRecord	Element		Yes	i4	11	Number of valid tickets	1099
BlacklistRecord	Element		Yes	i4	11	Number of blocked tickets Attribute BlockingType defines 0 …. Data Carrier Blocking 1 …. Permission Blocking	9

4.37.1 Totals sample

Request from TS

```
<Inquiry Type="Imports">
</Inquiry> fts
```

Answer from HSH

```
<Inquiry Type="Imports">
     <Totals>
        <WhitelistRecord>230</WhitelistRecord>
        <BlacklistRecord BlockingType="0">3</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
     </Totals>
</Inquiry>
```

4.38 ExtendedCheck element

Internal

The ExtendedCheck Element contains detailed information of extended checks like TwoStepValidation.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Name	Attribute		Yes	string	30	Unique name of extended check	TwoStep-Validation
EmployeeId	Attribute			I4	11	Unique Id of employee who executed the Extended Check	1
ResultCodes	Element			Elem.		ResultCodes of Extended Checks	

4.38.1 ResultCodes element

The ResultCodes element contains at least 1 ForeignCode element.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
ForeignCode	Element			Elem		See ForeignCode element	

4.38.1.1 ForeignCode element

The ForeignCode Element contains information if the ExtendedCheck was successful or not.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
Id	Attribute		Yes	I4	11	Unique id of ForeignCode	-2
Description	Attribute			String		Description of ForeignCode	<DetailedResult><BioscanResult OverallSuccess="Cancel" InfoMessage="action canceled"/> </DetailedResult>
ResultCodeId	Attribute			I4	11	Id of ResultCode	

Internal

4.38.1.2 Example for ExtendedCheck including ResultCodes

```
<ExtendedCheck Name="Biocheck V1.0">
     <ResultCodes>
        <ForeignCode Id="-2" Description="&lt;DetailedResult&gt;&lt;BioscanResult
                OverallSuccess=&quot;Cancel&quot; InfoMessage=&quot;action
                canceled&quot;/&gt;&lt;/DetailedResult&gt;"/>
     </ResultCodes>
</ExtendedCheck>
```

4.39 InquiryError element

The Inquiry Error contains information if the requested data does not exist or additional information for the returned data.

Name	Attribute / Element	Key	Mand.	Data Type	Max. Length	Remark	Example
No	Attribute	Yes	Yes	i4	11	Error Number	1000
Text	Attribute		Yes	string	100	Error Description	Unknown UPID

4.39.1 List of inquiry errors

The following table contains the Inquiry errors currently defined. This list may be changed and extended in the future.

No.	Error Number	Error Description	Remark
1	1000	Unknown UPID	Ticket does not Exist in Handshake™ database
2	1001	No Permission Type	No Permission Type (Day Ticket, Season Ticket, Time Ticket, ...) is configured for the specified ticket
3	1002	Time exceeded	Used time greater than purchased time
4	1003	No Time Ticket	TimeTicket Inquiry for none Time Ticket

4.40 Inquiry telegram samples

4.40.1 Inquiry of type Transactions

Request from TS

```
<Inquiry Type="Transactions" Option="128" From="2018-07-01T00:00:00" To="2018-12-01T20:00:00" />
```

Answer from HSH

```
<Inquiry Type="Transactions" Option="128" From="2018-07-01T00:00:00" To="2018-12-01T20:00:00">
     <Ticket>
        <UTID>30029659271</UTID>
        <Coding>262</Coding>
        <RawData>
           <DC>
              <L T="0">
                 <D T="6">30029659271</D>
              </L>
           </DC>
        </RawData>
        <Permission>
```


Internal

```
            <UPID>1</UPID>
            <Transaction TransactionId="1" Date="2018-07-01T08:30:21" Status="0" Validated ="0"
                     Releases="1">
               <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Area Type="1" Name="1" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Gate Name="Gate Reentry"/>
               <Checkpoint Name="Checkpoint 1"/>
            </Transaction>
            <Transaction TransactionId="21" Date="2018-08-03T11:00:21" Status="0" Validated ="0"
                     Releases="1">
               <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Area Type="1" Name="A1" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Gate Name="Gate Reentry"/>
               <Checkpoint Name="Checkpoint 1"/>
            </Transaction>
         </Permission>
      </Ticket>
      <Ticket>
         <UTID>30029659999</UTID>
         <Coding>262</Coding>
         <RawData>
            <DC>
               <L T="0">
                  <D T="6">30029659999</D>
               </L>
            </DC>
         </RawData>
         <Permission>
            <UPID>1</UPID>
            <Transaction TransactionId="13" Date="2018-07-02T08:30:21" Status="0" Validated ="0"
                     Releases="1">
               <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Area Type="1" Name="1" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Gate Name="Gate Reentry"/>
               <Checkpoint Name="Checkpoint 1"/>
            </Transaction>
            <Transaction TransactionId="14" Date="2018-08-04T11:00:21" Status="0" Validated="0"
                     Releases="1">
               <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Area Type="1" Name="A1" Entry="1" Exit="0" FE="0" FDE="1"/>
               <Gate Name="Gate Reentry"/>
               <Checkpoint Name="Checkpoint 1"/>
            </Transaction>
         </Permission>
      </Ticket>
</Inquiry>
```


Internal

4.40.2 Inquiry of type Transactions with option to return TSProperty names

Request from TS

```
<Inquiry Type="Transactions" Option="32991" LastTransactionId="0">
```

Answer from HSH

```
<Inquiry Type="Transactions" Option="32991" LastTransactionId="0">
     <Ticket>
        <UTID>A000063706</UTID>
        <Coding>261</Coding>
        <Permission>
           <UPID>4232013</UPID>
           <TSProperty Type="AREA">
              <ID>2980</ID>
           </TSProperty>
           <TSProperty Type="DISPLAYMESSAGE">
              <ID>VERROCCHIO 19360726</ID>
           </TSProperty>
           <TSProperty Type="EVENT">
              <ID>15123</ID>
           </TSProperty>
           <TSProperty Type="PERSONCATEGORY">
              <ID>114</ID>
           </TSProperty>
           <Transaction TransactionId="5991" Date="2018-10-13T17:30:28" Status="0" Validated="0"
                     Releases="1" />
        </Permission>
     </Ticket>
```

4.40.3 Inquiry of type Transactions – all transactions of a ticket

Request from TS

```
<Inquiry Type="Transactions">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Transactions">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <RawData>
        <DC>
           <L T="0">
              <D T="6">30029659271</D>
           </L>
        </DC>
     </RawData>
     <Permission>
        <UPID>1</UPID>
        <Transaction TransactionId="1" Date="2018-07-01T08:30:21" Status="0" Validated ="0"
              Releases="1">
           <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
           <Area Type="1" Name="A1" Entry="1" Exit="0" FE="0" FDE="1"/>
           <Gate Name="Gate Reentry"/>
           <Checkpoint Name="Checkpoint 1"/>
        </Transaction>
     </Permission>
</Inquiry>
```


Internal

4.40.4 Inquiry of type Transactions – all transactions for a single permission of a ticket (multiple permission ticket)

Request from TS

```
<Inquiry Type="Transactions">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Transactions">
        <UTID>30029659271</UTID>
        <Coding>262</Coding>
        <RawData>
           <DC>
              <L T="0">
                <D T="6">30029659271</D>
              </L>
           </DC>
        </RawData>
        <Permission>
           <UPID>1</UPID>
           <Transaction TransactionId="1" Date="2018-07-01T08:30:21" Status="0" Validated ="0"
Releases="1">
              <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
              <Area Type="1" Name="1" Entry="1" Exit="0" FE="0" FDE="1"/>
              <Gate Name="Gate Reentry"/>
              <Checkpoint Name="Checkpoint 1"/>
           </Transaction>
           <Transaction TransactionId="21" Date="2018-08-03T11:00:21" Status="0" Validated ="0"
Releases="1">
              <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
              <Area Type="1" Name="A1" Entry="1" Exit="0" FE="0" FDE="1"/>
              <Gate Name="Gate Reentry"/>
              <Checkpoint Name="Checkpoint 1"/>
           </Transaction>
     </Permission>
</Inquiry>
```

4.40.5 Inquiry of type Transactions – all transactions for a specified period

Request from TS

```
<Inquiry Type="Transactions" Option="128" From="2018-07-01T00:00:00" To="2018-07-31T20:00:00" />
```

Answer from HSH

```
<Inquiry Type="Transactions" Option="128" From="2018-07-01T00:00:00" To="2018-07-01T20:00:00">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <RawData>
        <DC>
           <L T="0">
              <D T="6">30029659271</D>
           </L>
        </DC>
     </RawData>
     <Permission>
        <UPID>1</UPID>
        <Transaction TransactionId="1" Date="2018-07-01T08:30:21" Status="0" Validated ="0"
Releases="1">
           <Venue Name="Venue 4" Entry="1" Exit="0" FE="0" FDE="1"/>
           <Area Type="1" Name="A1" Entry="1" Exit="0" FE="0" FDE="1"/>
           <Gate Name="Gate Reentry"/>
           <Checkpoint Name="Checkpoint 1"/>
        </Transaction>
     </Permission>
```


Internal

```
</Inquiry>
```

Note: UTID and Coding may be empty e.g. in case of a manual release (turnstile is opened using software or special tickets).

Sample:

```
<Inquiry Type="Transactions" Option="128" From="2018-07-01T00:00:00" To="2018-07-01T20:00:00">
     <UTID></UTID>
     <Coding></Coding>
     <Permission>
        <UPID>0</UPID>
        <Transaction TransactionId="1289" Date="2018-07-01T08:30:21" Status="62" Validated ="0"
                  Releases="1">
           <Venue Name="Venue 5" Entry="1" Exit="0" FE="0" FDE="1"/>
           <Area Type="1" Name="A3" Entry="1" Exit="0" FE="0" FDE="1"/>
           <Gate Name="Gate South 1"/>
           <Checkpoint Name="Checkpoint 7"/>
        </Transaction>
     </Permission>
</Inquiry>
```

4.40.6 Inquiry of type Transactions – suppress information with option element

In this example the elements area, gate and checkpoint are suppressed.

Request from TS

```
<Inquiry Type="Transactions" Option="14">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Transactions">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <RawData>
        <DC>
           <L T="0">
                <D T="6">30029659271</D>
           </L>
        </DC>
     </RawData>
     <Permission>
        <UPID>1</UPID>
        <Transaction TransactionId="1" Date="2018-07-01T08:30:21" Status="0" Validated ="0"
                Releases="1">
           <Venue Name="Legoland 4" Entry="1" Exit="0" FE="0" FDE="1"/>
        </Transaction>
        <Transaction TransactionId="2345" Date="2018-08-03T11:00:21" Status="0" Validated ="0"
                Releases="1">
           <Venue Name="Legoland 4" Entry="1" Exit="0" FE="0" FDE="1"/>
        </Transaction>
     </Permission>
</Inquiry>
```


Internal

4.40.7 Inquiry of type Transactions with conditions

Request from TS

```
<Inquiry Type="Transactions" Option="263199" Test="0" From="2017-06-07T15:14:42" />
```

Answer from HSH

```
<Inquiry Type="Transactions" Option="263199" Test="0" From="2017-06-07T15:14:42">
     <UTID>82719010418999</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>1</UPID>
        <TSProperty Type="AREA">
           <ID>10</ID>
        </TSProperty>
        <TSProperty Type="EVENT">
           <ID>1</ID>
        </TSProperty>
        <TSProperty Type="PERSONCATEGORY">
           <ID>19</ID>
        </TSProperty>
        <Transaction TransactionId="1" Date="2017-06-
                     07T16:14:42" Status="0" Validated="0" Releases="1" TestMode="0">
           <Conditions>
              <Condition ConditionId="1" ConditionName="Permit" ConditionType="0" />
              <Condition ConditionId="2" ConditionName="Photo capture" ConditionType="7" />
           </Conditions>
        </Transaction>
     </Permission>
</Inquiry>
```

4.40.8 Inquiry of type Permissions – all permissions of a ticket

Request from TS

```
<Inquiry Type="Permissions" Option="128" Status="All">
     <UTID>19681234</UTID>
     <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions" Option="128" Status="All">
     <Ticket Blocked="No">
        <UTID>19681234</UTID>
        <Coding>262</Coding>
        <NumberOfUPID>3</NumberOfUPID>
        <Permission Blocked="No">
           <UPID>1</UPID>
           <Expires>2018-11-06T01:00:00</Expires>
           <TSProperty Type="EVENT">
              <ID>1</ID>
           </TSProperty>
           <TSProperty Type="AREA">
              <ID>A1</ID>
           </TSProperty>
           <TSProperty Type="PERSONCATEGORY">
              <ID>A</ID>
           </TSProperty>
        </Permission>
        <Permission Blocked="No">
           <UPID>2</UPID>
           <Expires>2018-12-31T23:59:00</Expires>
           <TSProperty Type="AREA">
              <ID>A1</ID>
           </TSProperty>
           <TSProperty Type="PERSONCATEGORY">
              <ID>A</ID>
           </TSProperty>
           <TSProperty Type="EVENT">
              <ID>2</ID>
```


Internal

```
            </TSProperty>
         </Permission>
      </Ticket>
</Inquiry>
```

4.40.9 Inquiry of type Permissions – all tickets to a multi-data carrier permission

The element Permission is mandatory to receive the correct NumberOfUPID in case of a MultiDC.

Request from TS

```
<Inquiry Type="Permissions">
     <Permission MultiDC="true">
        <UPID>perm202_2</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions">
     <Ticket Blocked="No">
        <UTID>ticket202</UTID>
        <Coding>262</Coding>
        <NumberOfUPID>1</NumberOfUPID>
        <Permission Blocked="No" MultiDC="true">
           <UPID>perm202_2</UPID>
           <Expires>2017-12-31T01:02:03+04:00</Expires>
           <TSProperty Type="EVENT">
              <ID>2022</ID>
           </TSProperty>
        </Permission>
     </Ticket>
     <Ticket Blocked="No">
        <UTID>ticket203</UTID>
        <Coding>262</Coding>
        <NumberOfUPID>1</NumberOfUPID>
        <Permission Blocked="No" MultiDC="true">
           <UPID>perm202_2</UPID>
           <Expires>2017-12-31T01:02:03+04:00</Expires>
           <TSProperty Type="EVENT">
              <ID>2022</ID>
           </TSProperty>
        </Permission>
     </Ticket>
</Inquiry>
```

4.40.10 Inquiry of type Permissions – certain ticket to a multi-data carrier

The ticket can have multiple (1:n) permissions assigned. In this case it is only one permission. For this inquiry the user doesn’t know the permission hence UPID=-1 only the ticket (UTID and coding) is known.
The response includes the ticket itself and the assigned permission. Here it is a MultiDC permission.

Request from TS

```
<Inquiry Type="Permissions">
     <UTID>ticket202</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true">
        <UPID>-1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions">
```


Internal

```
    <Ticket Blocked="No">
       <UTID>ticket202</UTID>
       <Coding>262</Coding>
       <NumberOfUPID>1</NumberOfUPID>
       <Permission Blocked="No" MultiDC="true">
          <UPID>perm202_2</UPID>
          <Expires>2017-12-31T01:02:03+04:00</Expires>
          <TSProperty Type="EVENT">
             <ID>2022</ID>
          </TSProperty>
       </Permission>
    </Ticket>
</Inquiry>
```

4.40.11 Inquiry of type Permissions – ticket is blocked

The ticket has only one permission, which is a multi-data carrier permission.

Request from TS

```
<Inquiry Type="Permissions" Status="All">
    <UTID>ticket100</UTID>
    <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions" Status="All">
    <Ticket Blocked="Yes">
       <UTID>ticket100</UTID>
       <Coding>262</Coding>
       <NumberOfUPID>1</NumberOfUPID>
       <Permission Blocked="No" MultiDC="true">
          <UPID>perm100_1</UPID>
          <Expires>2017-12-31T01:02:03+04:00</Expires>
          <TSProperty Type="EVENT">
             <ID>1001</ID>
          </TSProperty>
       </Permission>
    </Ticket>
</Inquiry>
```

4.40.12 Inquiry of type Permissions – ticket not blocked, all permissions are blocked

The ticket has two permissions, which both are blocked. The inquiry response sets the Blocked attribute of Ticket XML-element to “No” because the ticket hasn’t been explicitly blocked. One of the permissions is a multi-data carrier permission, hence the MultiDC attribute in the Permission XML-element.

Request from TS

```
<Inquiry Type="Permissions" Status="All">
    <UTID>ticket101</UTID>
    <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions" Status="All">
    <Ticket Blocked="No">
       <UTID>ticket101</UTID>
       <Coding>262</Coding>
       <NumberOfUPID>2</NumberOfUPID>
       <Permission Blocked="Yes" MultiDC="true">
          <UPID>perm100_1</UPID>
          <Expires>2017-12-31T01:02:03+04:00</Expires>
          <TSProperty Type="EVENT">
```


Internal

```
            <ID>1001</ID>
         </TSProperty>
      </Permission>
      <Permission Blocked="Yes">
         <UPID>perm101_1</UPID>
         <Expires>2017-12-31T01:02:03+04:00</Expires>
         <TSProperty Type="EVENT">
            <ID>1011</ID>
         </TSProperty>
      </Permission>
   </Ticket>
</Inquiry>
```

4.40.13 Inquiry of type Permissions – all data

Request from TS

```
<Inquiry Type="Permissions">
   <UTID>30029659271</UTID>
   <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions">
   <Ticket>
      <UTID>30029659271</UTID>
      <Coding>262</Coding>
      <NumberOfUPID>6</NumberOfUPID>
      <Permission>
         <UPID>1</UPID>
         <TSProperty Type="EVENT">
            <ID>1</ID>
         </TSProperty>
         <TSProperty Type="AREA">
            <ID>A</ID>
         </TSProperty>
      </Permission>
      <Permission>
         <UPID>2</UPID>
         <TSProperty Type="EVENT">
            <ID>2</ID>
         </TSProperty>
         <TSProperty Type="AREA">
            <ID>A</ID>
         </TSProperty>
      </Permission>
      </Ticket>
</Inquiry>
```

4.40.14 Inquiry of type Permissions – suppress TSProperties:

Request from TS

```
<Inquiry Type="Permissions" Option="32">
   <UTID>30029659271</UTID>
   <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions" Option="32">
   <Ticket>
      <UTID>30029659271</UTID>
      <Coding>262</Coding>
      <NumberOfUPID>6</NumberOfUPID>
      <Permission>
         <UPID>1</UPID>
      </Permission>
      <Permission>
         <UPID>2</UPID>
      </Permission>
```


Internal

```
    </Ticket>
</Inquiry>
```

4.40.15 Inquiry of type Permissions – only number of existing permissions

Request from TS

```
<Inquiry Type="Permissions" Option="64">
<UTID>30029659271</UTID>
<Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions" Option="64">
    <Ticket>
        <UTID>30029659271</UTID>
        <Coding>262</Coding>
        <NumberOfUPID>17</NumberOfUPID>
    </Ticket>
</Inquiry>
```

4.40.16 Inquiry of type Permissions – all permissions using Status:

Request from TS

```
<Inquiry Type="Permissions" Option="128" Status="All">
    <UTID>30-16147115512941017248-5</UTID>
    <Coding>16777246</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Permissions" Option="128" Status="All">
    <Ticket Blocked="No">
        <UTID>30-16147115512941017248-5</UTID>
        <Coding>16777246</Coding>
        <NumberOfUPID>5</NumberOfUPID>
        <Permission Blocked="No">
            <UPID>HS2V8-99-07-138-1</UPID>
            <TSProperty Type="Product">
                <ID>LIP2</ID>
            </TSProperty>
            <TSProperty Type="POINTS">
                <ID>20</ID>
            </TSProperty>
        </Permission>
        <Permission Blocked="Yes">
            <UPID>HS2V8-99-07-138-2</UPID>
            <TSProperty Type="Product">
                <ID>LIP2</ID>
            </TSProperty>
            <TSProperty Type="POINTS">
                <ID>20</ID>
            </TSProperty>
        </Permission>
    </Ticket>
</Inquiry>
```

4.40.17 Inquiry of type Days with UTID, Coding and Permission

Request from TS

```
<Inquiry Type="Days">
    <UTID>ticket802</UTID>
    <Coding>262</Coding>
    <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                     is not a multi-data carrier permission -->
        <UPID>perm800_2</UPID>
    </Permission>
</Inquiry>
```


Internal

Answer from HSH

```
<Inquiry Type="Days">
     <UTID>ticket801</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                      is not a multi-data carrier permission -->
        <UPID>perm800_2</UPID>
        <Days Purchased="12" Used="0" />
     </Permission>
</Inquiry>
```

4.40.18 Inquiry of type Days with multi-data carrier permission

Request from TS

```
<Inquiry Type="Days">
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                      is not a multi-data carrier permission -->
        <UPID>perm800_2</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Days">
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                      is not a multi-data carrier permission -->
        <UPID>perm800_2</UPID>
        <Days Purchased="12" Used="0" />
     </Permission>
</Inquiry>
```

4.40.19 Inquiry of type Points with UTID, Coding and Permission

Request from TS

```
<Inquiry Type="Points">
     <UTID>ticket901</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                      is not a multi-data carrier permission -->
        <UPID>perm900_1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Points">
     <UTID>ticket901</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                      is not a multi-data carrier permission -->
        <UPID>perm900_1</UPID>
        <Points Purchased="10" Used="5" />
     </Permission>
</Inquiry>
```

4.40.20 Inquiry of type Points with multi-data carrier permission

Request from TS

```
<Inquiry Type="Points">
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                      is not a multi-data carrier permission -->
        <UPID>perm900_1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Points">
```


Internal

```
    <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                    is not a multi-data carrier permission -->
        <UPID>perm900_1</UPID>
        <Points Purchased="10" Used="5" />
    </Permission>
</Inquiry>
```

4.40.21 Inquiry of type Imports for the current day

Request from TS

```
<Inquiry Type="Imports" >
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Imports">
    <Totals>
        <WhitelistRecord>1024</WhitelistRecord>
        <BlacklistRecord BlockingType="0">9</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
    </Totals>
</Inquiry>
```

4.40.22 Inquiry of type Imports for a certain period

Request from TS

```
<Inquiry Type="Imports" From="2018-06-20T00:00:00" To="2018-06-21T12:30:00">
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Imports" From="2018-06-20T00:00:00" To="2018-06-21T12:30:00">
    <Totals>
        <WhitelistRecord>928</WhitelistRecord>
        <BlacklistRecord BlockingType="0">5</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
    </Totals>
/Inquiry>
```


Internal

4.40.23 Inquiry of type Imports for a certain Data Carrier Types (=Coding)

Request from TS

```
<Inquiry Type="Imports">
     <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Imports">
     <Coding>262</Coding>
     <Totals>
        <WhitelistRecord>357</WhitelistRecord>
        <BlacklistRecord BlockingType="0">3</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
     </Totals>
</Inquiry>
```

4.40.24 Inquiry of type Imports using Combination of all Options

Request from TS

```
<Inquiry Type="Imports" From="2018-06-20T00:00:00" To="2018-06-21T12:30:00">
     <Coding>262</Coding>
     <TSProperty Type="EVENT">
        <ID>01</ID>
     </TSProperty>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Imports" From="2018-06-20T00:00:00" To="2018-06-21T12:30:00">
     <Coding>262</Coding>
     <TSProperty Type="EVENT">
        <ID>01</ID>
     </TSProperty>
     <Totals>
        <WhitelistRecord>230</WhitelistRecord>
        <BlacklistRecord BlockingType="0">3</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
     </Totals>
</Inquiry>
```

4.40.25 Inquiry of type Imports with one TSProperty

Request from TS

```
<Inquiry Type="Imports" From="2018-01-01T00:00:00" To="2017-12-31T23:59:59">
     <TSProperty Type="EVENT">
        <ID>1</ID>
     </TSProperty>
</Inquiry>
```

or

```
<Inquiry Type="Imports" From="2018-01-01T00:00:00" To="2017-12-31T23:59:59">
     <TSProperties>
        <TSProperty Type="EVENT">
           <ID>1</ID>
        </TSProperty>
     </TSProperties>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Imports" From="2018-01-01T00:00:00" To="2017-12-31T23:59:59">
     <TSProperty Type="EVENT">
        <ID>1</ID>
     </TSProperty>
     <Totals>
        <WhitelistRecord>0</WhitelistRecord>
```


Internal

```
        <BlacklistRecord BlockingType="0">0</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
     </Totals>
</Inquiry>
```

4.40.26 Inquiry of type Imports with two TSProperties

Request from TS

```
<Inquiry Type="Imports" From="2017-01-01T00:00:00">
     <TSProperties>
        <TSProperty Type="EVENT">
           <ID>1</ID>
        </TSProperty>
        <TSProperty Type="AREA">
           <ID>10</ID>
        </TSProperty>
     </TSProperties>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Imports" From="2017-01-01T00:00:00">
     <TSProperties>
        <TSProperty Type="EVENT">
           <ID>1</ID>
        </TSProperty>
        <TSProperty Type="AREA">
           <ID>10</ID>
        </TSProperty>
     </TSProperties>
     <Totals>
        <WhitelistRecord>2</WhitelistRecord>
        <BlacklistRecord BlockingType="0">1</BlacklistRecord>
        <BlacklistRecord BlockingType="1">1</BlacklistRecord>
     </Totals>
</Inquiry>
```

4.40.27 Inquiry of type Imports with no TSProperties

Request from TS

```
<Inquiry Type="Imports"></Inquiry>
```

Answer from HSH

The result specifies all whitelist permission, blacklist permissions and blacklist tickets which have been received at the current day (the day when the inquiry has been sent).

```
<Inquiry Type="Imports">
     <Totals>
        <WhitelistRecord>1000</WhitelistRecord>
        <BlacklistRecord BlockingType="0">0</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
     </Totals>
</Inquiry>
```

4.40.28 Inquiry of type Imports with GroupBy

Request from TS

```
<Inquiry Type="Imports">
     <TSProperties>
        <TSProperty Type="EVENT">
           <ID>1</ID>
        </TSProperty>
        <TSProperty Type="AREA">
           <ID>10</ID>
        </TSProperty>
     </TSProperties>
     <GroupBy TSProperty="PERSONCATEGORY" />
```


Internal

```
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Imports">
     <TSProperties>
        <TSProperty Type="EVENT">
           <ID>1</ID>
        </TSProperty>
        <TSProperty Type="AREA">
           <ID>10</ID>
        </TSProperty>
     </TSProperties>
     <Totals>
        <TSProperty Type="PERSONCATEGORY">
           <ID>19</ID>
        </TSProperty>
        <WhitelistRecord>1</WhitelistRecord>
        <BlacklistRecord BlockingType="0">0</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
     </Totals>
     <Totals>
        <TSProperty Type="PERSONCATEGORY">
           <ID>20</ID>
        </TSProperty>
        <WhitelistRecord>1</WhitelistRecord>
        <BlacklistRecord BlockingType="0">0</BlacklistRecord>
        <BlacklistRecord BlockingType="1">0</BlacklistRecord>
     </Totals>
</Inquiry>
```

4.40.29 Inquiry of type TicketUsage

Request from TS:

```
<Inquiry Type="TicketUsage">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
</Inquiry>
```

Answer from HSH:

```
<Inquiry Type="TicketUsage">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <RawData>
        <DC>
           <L T="0">
              <D T="6">30029659271</D>
           </L>
        </DC>
     </RawData>
     <Permission>
        <UPID>1</UPID>
        <TicketUsed>1</TicketUsed>
        <LastDate>2018-03-01T12:25:33+04:00</LastDate>
     </Permission>
</Inquiry>
```

4.40.30 Inquiry of type TicketUsage with multi-data carrier permission

The inquiry request of type TicketUsage must specify the multi-data carrier permission.

Request from TS:

```
<Inquiry Type="TicketUsage">
     <UTID>Ticket1</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true">
        <UPID>1</UPID>
     </Permission>
</Inquiry>
```


Internal

Answer from HSH:

```
<Inquiry Type="TicketUsage">
     <UTID>Ticket1</UTID>
     <Coding>262</Coding>
     <RawData>
        <DC>
           <L T="0">
              <D T="6">Ticket1</D>
           </L>
        </DC>
     </RawData>
     <Permission MultiDC="true">
        <UPID>1</UPID>
        <TicketUsed>1</TicketUsed>
        <LastDate>2017-02-03T11:01:34+04:00</LastDate>
     </Permission>
</Inquiry>
```

4.40.31 Inquiry of type TicketUsage with TSProperties

Request from TS:

```
<Inquiry Type="TicketUsage">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH:

```
<Inquiry Type="TicketUsage">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <RawData>
        <DC>
           <L T="0">
              <D T="6">30029659271</D>
           </L>
        </DC>
     </RawData>
     <Permission>
        <UPID>1</UPID>
        <TicketUsed>1</TicketUsed>
        <LastDate>2018-03-01T00:00:00+04:00</LastDate>
     </Permission>
</Inquiry>
```

4.40.32 Inquiry of type TicketUsage – usage for a data carrier, ticket not used

Request from TS

```
<Inquiry Type="TicketUsage">
    <UTID>photoBRMI</UTID>
     <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="TicketUsage">
     <UTID>photoBRMI</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>-1</UPID>
        <TicketUsed>0</TicketUsed>
     </Permission>
</Inquiry>
```


Internal

4.40.33 Inquiry of type TicketUsage – usage for a data carrier, ticket unknown

Request from TS

```
<Inquiry Type="TicketUsage">
     <UTID>photoNIX</UTID>
     <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="TicketUsage">
     <UTID>photoNIX</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>-1</UPID>
        <InquiryError No="1000" Text="unknown UPID" />
     </Permission>
</Inquiry>
<Error Type="E">
     <Number>20014</Number>
     <Description>Non Existing Permission</Description>
     <Inquiry Type="TicketUsage">
        <UTID>photoNIX</UTID>
        <Coding>262</Coding>
     </Inquiry>
</Error>
```

4.40.34 Inquiry of type TimeTicketPeriod with UTID, Coding and Permission

Request from TS

```
<Inquiry Type="TimeTicketPeriod">
     <UTID>ticket701</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                  is not a multi-data carrier permission -->
        <UPID>perm700_1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="TimeTicketPeriod">
     <UTID>ticket701</UTID>
     <Coding>262</Coding>
     <Permission MultiDC="true"> <!-- or <Permission MultiDC="false"> or <Permission> if the permission
                                  is not a multi-data carrier permission -->
        <UPID>perm700_1</UPID>
        <Time Purchased="15" Used="12" />
     </Permission>
</Inquiry>
```

4.40.35 Inquiry of type TimeTicketPeriod with multi-data carrier permission

Request from TS

```
<Inquiry Type="TimeTicketPeriod">
     <Permission MultiDC="true">
        <UPID>perm700_1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="TimeTicketPeriod">
     <Permission MultiDC="true">
```


Internal

```
        <UPID>perm700_1</UPID>
        <Time Purchased="15" Used="12" />
    </Permission>
</Inquiry>
```

4.40.36 Inquiry of type TimeTicketPeriod

This sample shows an Inquiry used to receive the purchased and the used amount of time of a time ticket.

Request from TS

```
<Inquiry Type="TimeTicketPeriod">
    <UTID>30029659271</UTID>
    <Coding>262</Coding>
    <Permission>
        <UPID>1</UPID>
    </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="TimeTicketPeriod">
    <UTID>30029659271</UTID>
    <Coding>262</Coding>
    <Permission>
        <UPID>1</UPID>
        <Time Purchased="180" Used="30"/>
    </Permission>
</Inquiry>
```

4.40.37 Inquiry of type Information for certain Data Carrier

This sample shows an Inquiry used to receive the personalized data of a certain ticket.

Request from TS

```
<Inquiry Type="Information">
    <UTID>SD02.01.2017000001</UTID>
    <Coding>262</Coding>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Information">
  <PermissionInfo>
    <Ticket Blocked="Yes">
      <UTID>SD02.01.2017000001</UTID>
      <Coding>262</Coding>
      <NumberOfUPID>1</NumberOfUPID>
      <Permission Blocked="Yes">
        <UPID>1</UPID>
        <TSProperty Type="TicketId">
          <ID>000001</ID>
        </TSProperty>
        <TSProperty Type="Date">
          <ID>02.01.2017</ID>
        </TSProperty>
        <TSProperty Type="Days">
          <ID>2/5</ID>
        </TSProperty>
        <TSProperty Type="CDF">
          <ID>generic:2</ID>
        </TSProperty>
        <TSProperty Type="Stadium">
          <ID>SD</ID>
        </TSProperty>
      </Permission>
    </Ticket>
  </PermissionInfo>
  <TransactionInfo>
```


Internal

```
    <Transaction TransactionId="74" Date="2018-07-26T13:58:40" Status="0" Validated="0" Releases="1"
TestMode="0">
      <Venue Name="Venue" Entry="1" Exit="0" FE="1" FDE="1" />
      <Area Type="1" Name="Area A" Entry="1" Exit="0" FE="1" FDE="1" />
      <Gate Name="Gate Entry Area A 1" />
      <Checkpoint Name="VG Sim" />
    </Transaction>
    <Transaction TransactionId="75" Date="2018-07-26T13:58:48" Status="0" Validated="0" Releases="1"
TestMode="0">
      <Venue Name="Venue" Entry="1" Exit="0" FE="0" FDE="0" />
      <Area Type="1" Name="Area A" Entry="1" Exit="0" FE="0" FDE="0" />
      <Gate Name="Gate Entry Area A 1" />
      <Checkpoint Name="VG Sim" />
    </Transaction>
  </TransactionInfo>
  <ForwardedData>
    <Requester TSId="1">
      <RequestedData Type="Permissions" Timestamp="2018-07-27T11:48:09" />
      <RequestedData Type="Permissions" Timestamp="2018-07-27T11:50:21" />
      <RequestedData Type="Permissions" Timestamp="2018-07-27T12:00:07" />
    </Requester>
  </ForwardedData>
</Inquiry>
```

4.40.38 Inquiry of type Information for certain Data Carrier and Permission

This sample shows an Inquiry used to receive the personalized data of a certain permission.

Request from TS

```
<Inquiry Type="Information">
     <UTID>SD02.01.2017000001</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Information">
  <PermissionInfo>
    <Ticket Blocked="Yes">
      <UTID>SD02.01.2017000001</UTID>
      <Coding>262</Coding>
      <NumberOfUPID>1</NumberOfUPID>
      <Permission Blocked="Yes">
        <UPID>1</UPID>
        <TSProperty Type="TicketId">
          <ID>000001</ID>
        </TSProperty>
        <TSProperty Type="Date">
          <ID>02.01.2017</ID>
        </TSProperty>
        <TSProperty Type="Days">
          <ID>2/5</ID>
        </TSProperty>
        <TSProperty Type="CDF">
          <ID>generic:2</ID>
        </TSProperty>
        <TSProperty Type="Stadium">
          <ID>SD</ID>
        </TSProperty>
      </Permission>
    </Ticket>
  </PermissionInfo>
  <TransactionInfo>
    <Transaction TransactionId="74" Date="2018-07-26T13:58:40" Status="0" Validated="0" Releases="1"
TestMode="0">
      <Venue Name="Venue" Entry="1" Exit="0" FE="1" FDE="1" />
      <Area Type="1" Name="Area A" Entry="1" Exit="0" FE="1" FDE="1" />
```


Internal

```
      <Gate Name="Gate Entry Area A 1" />
      <Checkpoint Name="VG Sim" />
    </Transaction>
    <Transaction TransactionId="75" Date="2018-07-26T13:58:48" Status="0" Validated="0" Releases="1"
TestMode="0">
      <Venue Name="Venue" Entry="1" Exit="0" FE="0" FDE="0" />
      <Area Type="1" Name="Area A" Entry="1" Exit="0" FE="0" FDE="0" />
      <Gate Name="Gate Entry Area A 1" />
      <Checkpoint Name="VG Sim" />
    </Transaction>
  </TransactionInfo>
  <ForwardedData>
    <Requester TSId="1">
      <RequestedData Type="Permissions" Timestamp="2018-07-27T11:48:09" />
      <RequestedData Type="Permissions" Timestamp="2018-07-27T11:50:21" />
      <RequestedData Type="Permissions" Timestamp="2018-07-27T12:00:07" />
    </Requester>
  </ForwardedData>
</Inquiry>
```

4.40.39 Inquiry of type Information for TSProperty

This sample shows an Inquiry used to receive the personalized data of a certain TSProperty.

Request from TS

```
<Inquiry Type="Information">
     <TSProperty Type="TicketId">
        <ID> 00000001</ID>
     </TSProperty>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Information">
  <PermissionInfo>
    <Ticket Blocked="Yes">
      <UTID>SD02.01.2017000001</UTID>
      <Coding>262</Coding>
      <NumberOfUPID>1</NumberOfUPID>
      <Permission Blocked="Yes">
        <UPID>1</UPID>
        <TSProperty Type="TicketId">
          <ID>000001</ID>
        </TSProperty>
        <TSProperty Type="Date">
          <ID>02.01.2017</ID>
        </TSProperty>
        <TSProperty Type="Days">
          <ID>2/5</ID>
        </TSProperty>
        <TSProperty Type="CDF">
          <ID>generic:2</ID>
        </TSProperty>
        <TSProperty Type="Stadium">
          <ID>SD</ID>
        </TSProperty>
      </Permission>
    </Ticket>
  </PermissionInfo>
  <TransactionInfo>
    <Transaction TransactionId="74" Date="2018-07-26T13:58:40" Status="0" Validated="0" Releases="1"
TestMode="0">
      <Venue Name="Venue" Entry="1" Exit="0" FE="1" FDE="1" />
      <Area Type="1" Name="Area A" Entry="1" Exit="0" FE="1" FDE="1" />
      <Gate Name="Gate Entry Area A 1" />
      <Checkpoint Name="VG Sim" />
    </Transaction>
    <Transaction TransactionId="75" Date="2018-07-26T13:58:48" Status="0" Validated="0" Releases="1"
TestMode="0">
      <Venue Name="Venue" Entry="1" Exit="0" FE="0" FDE="0" />
      <Area Type="1" Name="Area A" Entry="1" Exit="0" FE="0" FDE="0" />
      <Gate Name="Gate Entry Area A 1" />
```


Internal

```
      <Checkpoint Name="VG Sim" />
    </Transaction>
  </TransactionInfo>
  <ForwardedData>
    <Requester TSId="1">
      <RequestedData Type="Permissions" Timestamp="2018-07-27T11:48:09" />
      <RequestedData Type="Permissions" Timestamp="2018-07-27T11:50:21" />
      <RequestedData Type="Permissions" Timestamp="2018-07-27T12:00:07" />
    </Requester>
  </ForwardedData>
</Inquiry>
```

4.40.40 Inquiry of type Information for a MultiDC ticket

This sample shows an Inquiry used to receive the personalized data of a certain MultiDC ticket.

Request from TS

```
<Inquiry Type="Information">
     <Permission MultiDC="true">
        <UPID>12345</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Information">
     <PermissionInfo>
        ...
     </PermissionInfo>
     <TransactionInfo>
        ...
     </TransactionInfo>
     <ForwardedData>
        ...
     </ForwardedData>
</Inquiry>
```

4.40.41 Inquiry answered with error element

In this sample an error is returned from the HSH because the UPID is unknown to the HSH.

Request from TS

```
<Inquiry Type="Transactions" Option="14">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</Inquiry>
```

Answer from HSH

```
<Inquiry Type="Transactions">
     <UTID>30029659271</UTID>
     <Coding>262</Coding>
     <Permission>
        <UPID>1</UPID>
        <InquiryError No="1000" Text="unknown UPID"/>
     </Permission>
</Inquiry>
<Error Type="E">
     <Number>20014</Number>
     <Description>Non Existing Permission</Description>
     <Inquiry Type="Transactions">
        <UTID>30029659271</UTID>
        <Coding>262</Coding>
     </Inquiry>
</Error>
```


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5. Coding definition table

The Coding information is stored within 32 Bits (4 Bytes). The single bits are grouped together to save the required information. The meaning of the bits differs depending on the used technology barcode or radio frequency. All bits marked as FFU (for future use) are not used now and must be set to zero.

FFU		Coding technology						FFU								Technology Options															
31	30	29					24	23							16	15							8	7							0

5.1 Coding technology (Bits 24-29)

Bit 24-29 defines the used coding technology.

Value	Technology
0	Barcode
1	Radio Frequency Technology (RF)
2	Special Codings
4	License Plate (LPR)

Depending on the selected coding technology, the technology options use values as described below.

5.1.1 Coding technology barcode

FFU		Coding technology						FFU								Technology Options															
31	30	29					24	23							16	15		9	8	7											0
		Barcode														FFU			Visible	Coding Types												

5.1.1.1 Barcode coding types

Bit 0-7 of the technology options contains the information about the barcode type.

Value	Barcode coding type
0	Standard 2 of 5
1	Interleaved 2 of 5
2	UPC
3	PDF417, Macro PDF417, Micro PDF417 (2D)
4	EAN
5	Code 39

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6	Code 128
7	Codabar
11	DataMatrix (2D)
14	QR Code
18	Aztec

5.1.1.2 Barcode visibility options

Bit 8 of the technology options defines, if the barcode is coded visible or invisible (infrared)

Value	Barcode visibility option
0	Invisible barcode (920 nm, infrared) – Note: not used anymore
1	Visible barcode (640 nm)

5.1.2 Coding technology RF

FFU		Coding technology							FFU							Technology Options											
31	30	29					24	23							16	15		8	7								0
		Radio Frequency														FFU			RF Chip ID								

5.1.2.1 RF Chip ID

Bit 0-7 of the technology options contains the information about the RF chip ID.

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Value	RF chip ID	Frequency	Comment
1	Flexspace (ISO 14443/ ISO 15693)	13,56 MHz	
6	HID iCLASS (ISO 14443/ ISO 15693)	13,56 MHz	Serial Number only
7	Felica Chip (ISO 14443)	13,56 MHz	Serial Number only
13	Generic ISO 15693	13,56 MHz	Serial Number only
15	Calypso	13,56 MHz	Serial Number only
17	ISO 14443 Portugal – ASK	13,56 MHz	
19	NFC generic	13,56 MHz	
21	EM4036 Shared Chip (ISO 15693)	13,56 MHz	
25	Keycard ECO ISO DUAL	13,56 MHz	
26	Keyticket ISO DUAL	13,56 MHz	
29	Keycard ISO (ISO 15693)	13,56 MHz	
30	Keycard ISO DUAL (ISO 15693)	13,56 MHz	
132	Mifare/MyD (ISO 14443A)	13,56 MHz	
133	Mifare (ISO 14443B)	13,56 MHz	

5.1.3 Special coding

FFU		Coding technology						FFU								Technology Options							
31	30	29					24	23							16	15		5	4				0
		Special Coding														FFU			Special Coding Type				

5.1.3.1 Special coding types

Bit 0-4 of the technology options defines special coding. The value 0 is used if the Coding information cannot be provided by the TS.

Value	Special Coding Types
0	Coding not defined
1-15	FFU

5.1.4 Coding technology license plate (LPR)

FFU		Coding technology						FFU								Technology Options							
31	30	29					24	23							16	15		7	6				0
		License Plate (LPR)														FFU			License Plate Type				

Internal

5.1.4.1 License plate types

Bit 0-6 of the technology options defines the license plate type. The value 0 is typically used unless the TS requires a differentiation of license plates with the same content.

Value	License Plate Types
0	License Plate
1-99	FFU

5.1.5 Coding technology Biometric

Introduced to be used for Face Recognition. The coding information holds the ID provided by the ticket partner to be used as backup if the Biometric look up do not work. This can be the RF card serial number.

5.2 Samples for valid coding information

Value	Technology	Options	Coding Type Data Carrier
256	Barcode	Visible	Standard 2 of 5
257	Barcode	Visible	Interleaved 2 of 5
258	Barcode	Visible	UPC
259	Barcode	Visible	PDF417, Macro PDF417, Micro PDF417 (2D)
260	Barcode	Visible	EAN
261	Barcode	Visible	Code 39
262	Barcode	Visible	Code 128
263	Barcode	Visible	Codabar
267	Barcode	Visible	DataMatrix (2D)
270	Barcode	Visible	QR Code (2D)
274	Barcode	Visible	Aztec (2D)
16777217	RF		Flexspace
16777222	RF		HID iCLASS
16777223	RF		Felica RFID
16777229	RF		Generic ISO 15693
16777231	RF		Calypso
16777235	RF		NFC generic
16777237	RF		EM4036 Shared Chip
16777241	RF		Keycard ECO ISO DUAL

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16777242	RF	Keyticket ISO DUAL
16777245	RF	Keycard ISO
16777246	RF	Keycard ISO DUAL
16777348	RF	Mifare/MyD/Ultralight (ISO 14443A)
16777349	RF	Mifare (ISO 14443B)
33554432	Special Coding	Coding not defined
67108864	License Plate (LPR)	License Plate
67108880	Biometric	Face Recognition

The shown list contains all combinations that are possible now. In the future the list may be extended when new technologies are implemented in the HSH-System.

6. Barcode content

Only few data can be stored on a barcode ticket. Therefore, it is important which information is coded on the ticket. Handshake.Logic is designed to work with existing barcode tickets but if new tickets are implemented the following ideas should be taken into consideration. In general, as much data as possible should be stored on the ticket. For the access system it's helpful if the following data is stored on the ticket especially if the HSH server is not available.

A definable amount of this information must build up the unique TicketId (UTID). The UTID guarantees the uniqueness of the ticket.

The following list of data potentially coded on a ticket shall give an idea what kind of information can be meaningful for HSH.

6.1 Barcode data

Field	Remark
Issuer	Unique Id for ticket producer
SystemId	Unique System ID
EventId / SeasonId	Unique Id for Event / Season
TicketId	Ticket Number
AreaId	Id for Area allowed to enter
PersonCategoryId	Person type (e.g.: adult, student)

6.1.1 Issuer

The Issuer specifies the producer of ticket (permission).

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Note: The Issuer in the barcode and the Issuer in XML-Header have different meanings and must be distinguished.

6.1.2 SystemId

The SystemId identifies the system itself, it can be used to distinguish between different HSH installations.

6.1.3 EventId / SeasonId

Each event resp. season receives a unique Id. Only tickets with the correct EventId or SeasonId can be accepted at the entrance.

6.1.4 TicketId (TID)

The TicketId is a unique number (in most cases starting number 1 and incremented by 1) coded on the ticket. The TicketId is only unique within the system and it may be unique only for one event.

6.1.5 AreaId

With the AreaId it's possible to allow or deny tickets into areas. If this information is stored on the ticket it is possible to check if the ticket is authorized to enter a certain area even if the checkpoints are working in offline mode.

6.1.6 PersonCategoryId

Besides the configuration of permissions and restrictions the PersonCategoryId can be used to control the light signals at the checkpoints. The light signals can be used to give optical information about the PersonCategory of tickets, e.g. Reduced Ticket green flashing.

7. RF data carrier

RF data carriers are mainly used for long term tickets like season passes, staff tickets etc. In most cases the same information used within the barcode is coded on the RF data carrier. Another way to use RF data carrier is to identify the ticket by the chip's serial number. In this case the RF Data Carrier is read-only, so the ticketing system doesn't need to encode the RF data carrier.

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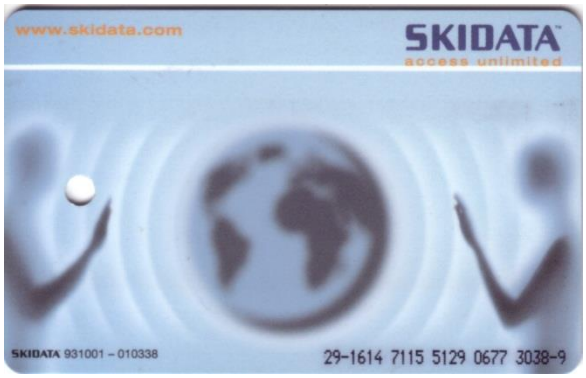
8. Read-only mode

Each RF data carrier has a unique serial number. In read-only mode this serial number is used to identify the ticket. The properties of the ticket are transferred within the Whitelist. In read-only mode the chip's serial number is the UTID. The read-only mode can be used for e.g. Internet booking. The customer selects the kind of ticket he wants and afterwards he must key in the serial number of his RF data carrier. The serial number is printed on the backside of the keycard or swatch. This serial number is matched to the existing WhitelistRecord and the combination between serial number and Whitelist information is the ticket type.

8.1.1 13 MHz Keycard



8.1.1.1 Example

```
<WhitelistRecord Expire="2018-12-31T00:00:00+04:00">
     <Action>I</Action>
     <UTID>16147115512906773038</UTID>
     <Coding>16777245</Coding>
     <Permission>
        <UPID>1</UPID>
     </Permission>
</WhitelistRecord>
```


Internal

9. E-ticket integration

9.1 Supported TSI requests

Request	Supported	Comment
Whitelist	Yes	The whitelist record is the source of e-ticket creation, update, delete and anonymize operations.
Blacklist	No	The blacklist record is not handled by TEG. It's interpreted as affecting HSH access control (permission granted y/n) only.
TSProperty	No	The TSProperty request is not handled by TEG. If the <Name> element of a TSProperty is used as data source for an e-ticket then it's assumed that the TS always sends the <Name> element in the whitelist record -> permission -> TSProperty.
Inquiry	No	A HSH inquiry is passed on unchanged and unprocessed to HSH. (ET inquiry is not implemented yet.)

9.2 Supported Whitelist actions

Action	Supported	Comment
Insert	Yes	The whitelist ***Insert*** action is transformed into an ET CreateTicket… (or UpdateTicket…) operation.
Update	Yes	The whitelist ***Update*** action is transformed into an ET UpdateTicket… (or CreateTicket…) operation.
Delete	Yes	The whitelist ***Delete*** operation is transformed into the voiding/cancellation the corresponding e-ticket.
Anonymize	Yes	The whitelist ***Anonymize*** operation is transformed into ET UnregisterUser… operations.

9.2.1 Automatic action adaption

TEG implement the same automatic adaption of the ***Insert*** and ***Update*** action as HSH:

- If an e-ticket is to be inserted (created) but already exists, the existing e-ticket is updated.
- If an e-ticket is to be updated but doesn't exist, a new e-ticket is inserted (created).

9.3 Supported data transfer

The data required for an e-ticket operation is mainly derived from the TEG configuration.

The TEG configuration specifies how an e-ticket property can be found in and derived from the TSI data. In addition, there is the option of defining default values in the TEG configuration in case the data is not available in TSI.

The TEG configuration is described in detail in the TEG manual and API documentation.

9.3.1 E-ticket creation

e-ticket data	Supported	Derived from / constructed by
UTID	Yes	TS ticket UTID and/or UPID

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ExternalId	Yes	ExternalIdProperty
DeliveryMethod	Yes	DeliveryMethodProperty
TicketDefinition	Yes	TicketDefinitionProperty
ServicePartner	Yes	ServicePartnerProperty
EndpointURI	Yes	EmailProperty
RelevantDate	Yes	RelevantDateProperty
ExpirationDate	Yes	WhitelistRecord.Expiry or DefaultExpiryDate
IdentificationType	Yes	CodingTranslation
SecondaryIdentificationType	Yes	CodingTranslation
DataCarrierContent	Yes	PrimaryDataCarrierContentProperty or TS ticket UTID
SecondaryDataCarrierContent	Yes	SecondaryDataCarrierContentProperty or TS ticket UTID
Photo	Yes	Whitelist.Permission.TSProperty of type “Photo”
Properties	Yes	PropertyTranslation
Locations	No	
Beacons	No	
AccessConfig	No	
DownloadLinkType	No	
CustomerAuthenticationInformation	No	

9.3.2 E-ticket update

e-ticket data	Supported	Derived from / constructed by
EndpointURI	Yes	EmailProperty
DataCarrierContent	Yes(1)	PrimaryDataCarrierContentProperty, only if PrimaryDataCarrierProperty is delivered in TS data
SecondaryDataCarrierContent	Yes(1)	SecondaryDataCarrierContentProperty,, only if PrimaryDataCarrierProperty is delivered in TS data
Photo	Yes	Whitelist.Permission.TSProperty of type “Photo”
RelevantDate	Yes	RelevantDateProperty

Internal

ExpirationDate	Yes	WhitelistRecord.Expiry or DefaultExpiryDate
Properties	Yes	PropertyTranslation
Voided	No	
IdentificationType	No	
SecondaryIdentificationType	No	
Locations	No	
Beacons	No	
CustomerAuthenticationInformation	No	

9.3.3 Reporting to TS

See 4.4.3 and 4.4.4 for the list of TEG warning and error codes and their usage in the response to the ticketing system.

9.3.4 Reporting to HSH

The status of an e-ticket operation is reported in a dedicated, predefined TSProperty - see 4.23 for details.

9.3.5 E-ticket download URL

If the ticketing system requests the download link for a created e-ticket (delivery by TS not by ET) then the URL is reported to the TS – see 4.4.2 for details.

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